01/24

Friday, 24 January 2025 9:30 AM

Direct Methods for $Ax = b$

+ Cramer's rule
+ Gaussian Elimination

  $Ax = b$ → converted to $Lx = b'$ or $Ux = b'$

+ Gauss-Jordon
+ LU decomposition

Variations of the same

When do you not need to do this?
↳ Diagonally dominant & positive definite

+ Full & partial pivoting [moving rows & columns]
+ Equilibration → normalize all eqn so you can compare the eqn
  → essential to do before pivoting

(direct)
These methods are memory intensive
(i.e. all the elements (even zero) need to be stored ↳ since they may even become non-zero)

Thomas Algorithm

Multiplying two diagonal matrices → gives tridiagonal matrix

Iterative Methods

+ Jacobi (parallelizable)
+ Gauss Seide (cannot be parallelized)

guaranteed always converge if A is diagonally dominant

How to check?
+ rearrange eqn so you have largest terms in diagonal
+ solve for just the diagonal elements

For linear systems → initial guess does not matter
↳ If no converge → successive over/under relaxation (SOR)
↳ introduce ω parameter

★

$Ax = b$ error analysis

+ Residuals
  ↳ But a small residual only makes sense if the matrix is ill-conditioned

Always do relative error!

$$\kappa(A) \geq 1$$

$\kappa(A)$ expensive to calculate

$$\kappa(A) = \|A^{-1}\| \|A\|$$

$$\frac{\|\delta x\|}{\|(x + \delta x)\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A\|}$$

+ Iterative Refinement Algorithm
  ↳ Calculate residual at twice the precision (significant figures)

(only one iteration p much needed)

01/29

Wednesday, 29 January 2025 9:31 AM

Midterms → only conceptual qs

Final → algorithm

Roots:

What method can guarantee root?

↳ Bisection (but have to cross axis i.e. change in value from + to −)

↳ Newton-Rhapson (N-R) (initial guess must be w/in convergence limit)

↳ If slope is zero (it will eventually approach it but never reach)

↳ Modified N.R. to deal w/ this

Ax=b

- Why pivoting?
  - ↳ Avoid division by zero & avoid floating / round-up error [trying to find the biggest number]
  - ↳ equilibration → division by biggest term in equation ↓ to compare each equation
- Checking solution: residual norm & iterative

- Diverging solutions for J & G-S methods?
  - ↳ Expand radius of convergence

PDE's

- ↳ Linear, almost linear, quasi-linear, non-linear
  - Highest order must be non-linear
  - Linear: when a no-directive term is non-linear
  - Quasi-linear: higher-order is linear but lower order is non-linear

- ↳ Second order PDE classification
  - + Second order coefficients matrix
  - + 5 different forms: parabolic, hyperbolic, elliptical
    - parabolic: narrow, broad

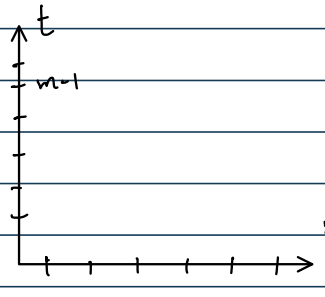
$U_t \Rightarrow \frac{\partial u}{\partial t}$

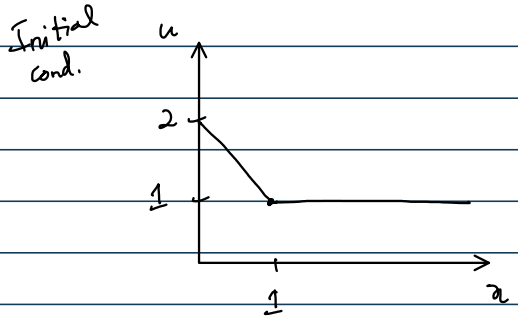
$UU_x \Rightarrow \frac{\partial u^2}{\partial x^2}$

Conservative vs. non-conservative form

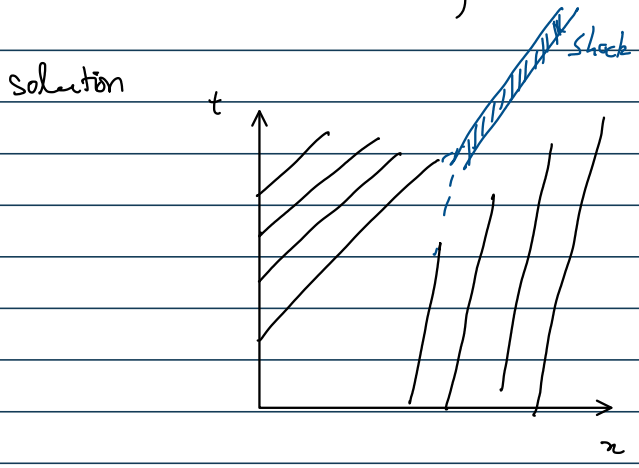
$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$$

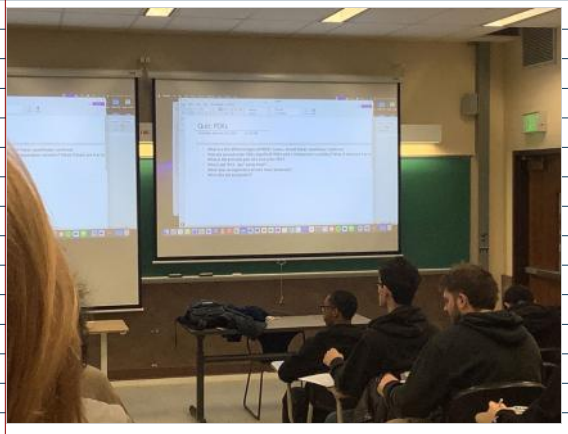
IC: $u \begin{cases} 2-x & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$





Multiple solns
Create "shock waves"
↓
discontinuity
(weak solution)

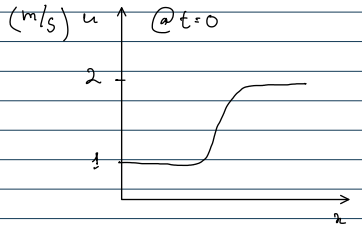




$$\frac{\partial u}{\partial t} + \underbrace{u\frac{du}{dx}}_{\frac{\partial}{\partial x}Q(u)} = 0$$

$$Q = \frac{u^2}{2} \quad ; \quad u_f = \frac{Q(u_2) - Q(u_1)}{u_2 - u_1}$$

Geometry, operating conditions

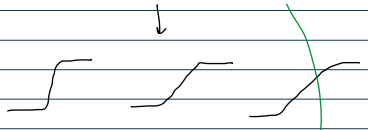


Given this, is this going to have weak solution? (for $t \to \infty$)

↓

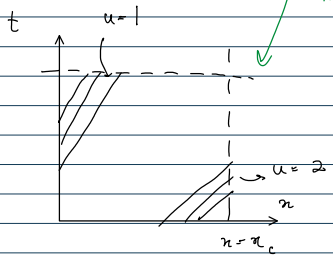
No, because it can't catch up! u is ???

vs.



Solutions through characteristic lines.

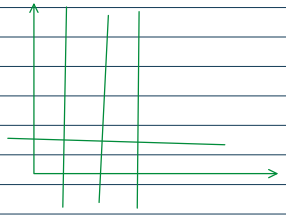
BC's : $u(x=0,t) = 1$



The lines can be drawn directing from this

↓

These are exact solutions.



$$\frac{(u_i^{n+1} - u_i^n)}{\Delta t} + \frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} = 0 \quad \Big\} \text{ explicit}$$

$$\frac{(u_i^{n+1} - u_i^n)}{\Delta t} + \frac{(u_i^{n+1})^2 - (u_{i-1}^{n+1})^2}{2\Delta x} = 0 \quad \Big\} \text{ implicit solutions.}$$

Numerical BC's

$$\frac{\partial^2 u}{\partial x^2} = 0$$

Second order PDE's :

- Elliptic → every point affects everything else (steady state potential flow)
- Hyperbolic → propagating solution up (allows weak solns)
- Parabolic
  - ↓ normal
  - ↳ broad
  - (one zero eigenvalue)

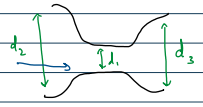
  } must specify BC at every point

Q3 (HW1) → no refinement
(discretize PDE's, BC's, system of non-linear eq's)



1.) Criteria for continuum

$$K_n = \frac{\text{mean free path}}{\text{characteristic length}} \lesssim 0.0001$$



might have to look at various BL's

2) Thermodynamic equilibrium

Reversibility, assume no gradients (uniformity) → things are infinitely slow

Why need thermo? →
→ define fluid properties (what are measurables)

} Can assume local equilibrium

→ Conc. + temp grad } coupling effects (one affecting the other)

3. Mechanical energy

02/05

Wednesday, 5 February 2025 9:30 AM

FD / FV

Finite Differences / finite element

approximates derivatives at nodes in a grid

control volume analysis (dividing up domain in control volume & integrating equations over these volumes)

Infinite no. of possible FD/FV eqs for given PDE

U_i^n → exact soln of PDE at (x_i, t^n)

u_i^n → exact soln of FDE/FVE at (x_i, t^n) ← no round off errors

N_i^n = numerical soln of FDE / FVE

Error at (x_i, t^n)

$$E_i^n = U_i^n - N_i^n = \underbrace{(U_i^n - u_i^n)}_{\text{discretization error}} + \underbrace{(u_i^n - N_i^n)}_{\text{round off / Stability crit}}$$

Consistency vs. Convergence vs. Stability

"identical twins"

(looks the same but does not mean convergence or stable) → Have to just do Taylor series expansion

Discretizing a PDE: → Han

If there is a ratio $\left(\frac{\Delta t}{\Delta x}\right)$ → then it will be conditionally stable

Stability Analysis

+ Discrete perturbation → Put a disturbance in soln. at some time & see its growth

+ Fourier (von Neumann) → disturbance at every grid point

+ Matrix

+ Modified eqn.

Fourier: $\varepsilon(x,t) = \frac{a_0}{2} + \sum (a_j \cos k_j + b_j \sin k_j x)$ $\quad k_j = \frac{2\pi j}{L}$

$\cos\psi = \frac{(e^{i\psi} + e^{-i\psi})}{2}$; $\sin\psi = \frac{(e^{i\psi} - e^{-i\psi})}{2i}$

Fourier; $f(x,t) = \frac{a_0}{2} + \sum (a_j \cos k_j x + b_j \sin k_j x)$ $\quad k_j = \frac{2\pi j}{L}$

+ Modified eqn.

$\cos\psi = \frac{(e^{i\psi} + e^{-i\psi})}{2}$; $\sin\psi = \frac{(e^{I\psi} - e^{-I\psi})}{2I}$

↳ But this must also be discretized
(there is an associated error for this as well)

→ Unconditional vs. conditional

$$\varepsilon_i = \sum_{j=-N}^{N} A_j^n \exp(I k_j x_i) \qquad \varepsilon_{i,t}^{n+1} = \sum A_j^{n+1} \exp$$

+ As you refine Δx & ΔT

$\frac{\Delta t}{\Delta x}$ } conditional (when discretized)

02/07

Friday, 7 February 2025 9:30 AM

Stability Analysis (contd.)

- Matrix method (accounting for boundary and as well as grid points)

$$T^{n+1} = AT^n + b$$

$$\varepsilon^{n+1} = A^{n+1}\varepsilon^0$$

ε^{n+1}: disturbance

A^{n+1}: depends on this coefficient.

A matrix is convergent if the spectral radius of it is less than 1

↳ multiplying matrix by itself again & again.

finding eigenvalues?

Gershgorin circle theorem

02/12

Wednesday, 12 February 2025 9:39 AM

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$



When does 2nd order derivative represent

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \mu\frac{\partial^2 u}{\partial x^2}$$

Even order → diffusion

odd order → derivatives } amplitude only depends on the

$$\frac{u_i^{n+1} - u_i^n}{\Delta x} + c\frac{u^{n+1} - u^n}{\Delta x} = \mu\left(\frac{u^{n+1} - u^n + u^{n-1}}{(\Delta x)^2}\right)$$

Shortest wavelength is the issue ...

Modified equation method

Courant number & CFL condition

↓

$= 1$

(one grid space in 1 time step)

↳ domain of influence of FD and ODE

Non-linear problem :

$$\frac{\partial T}{\partial t} = \alpha' \frac{\partial^2 T^{(m)}}{\partial x^2} \qquad m > 1$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha' \left(T_{i+1}^n\right)^{(m)} - 2\left(T_i^n\right)^{(m)}$$

$$F\left(T_{i\pm1}^n\right) = F\left(T_i^n\right) \pm \frac{\partial F\left(T_i^n\right)}{\partial x}\Delta x + O(\Delta x^2)$$ } First order approx.

↓

$$\frac{\partial F}{\partial T}\frac{\partial T}{\partial x}\Delta x \rightarrow \frac{\partial F}{\partial T}\left(\frac{T_{i+1} - T_i}{\Delta} + O(\Delta x)\right)\Delta x$$

$$\frac{\partial F}{\partial T}\frac{\partial T}{\partial x}\Delta x \rightarrow \frac{\partial F}{\partial T}\left(\frac{T_{i+1}-T_i}{\Delta x}+O(\Delta x)\right)\Delta x$$

$$=\left(\frac{\partial F}{\partial T}\right)(T_{i+1}-T_i)$$

$$\uparrow$$

$$F = T^{(m)}$$

02/17

Monday, 17 February 2025 9:35 AM

→ Stability of physics → PDE

→ But the discretization has its own stability → this is what we are concerned about.

Consistency	Convergence	Stability
As $\Delta x \to 0$, the solution looks similar to PDE	As $\Delta x \to 0$, the solution is the exact soln. of PDE	

↳ Can only prove for linear problems

↳ For non-linear can only linearize the check stability locally

↳ If the errors are bounded/ contained locally then don't care about the non-linearity

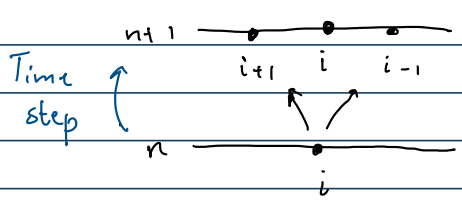
Checking the engineering:

1) Change to finite volume problem

(In - out

2) Transportive error - a disturbance is propagated (discrete perturbation analysis)

↳ error is felt beyond the solution



· Central differencing is adequate

3) Dissipation error - If the highest time derivative is one, then even order derivatives represent diffusion (momentum) or dissipation (mech. energy to thermal energy)

↳ Hard to avoid



G_{VW} → upp wind

↳ Shortest wavelength have higher dissipation

↳ Lax - Wendroff

Dispersive medium → waves w/ different wavelengths travel at different speeds in that medium

Non dispersive medium → all waves travel at the same speed

Wtf is a wave???

Dissipation ≠ Dispersion

↳ wher order is three

Review

Monday, 17 February 2025 12:35 PM

Linear Algebra

$$A\vec{x} = \vec{b}$$

- Unique solution exists if A is non-singular (i.e. if $\det(A) \neq 0$)

Eigenvalues of N×N matrix : $\det(A - \lambda I) = 0$; $A x_i = \lambda_i x_i$ (x_i: eigenvector)

- There are N eigenvalues ; depending on A they might be real or complex (i.e. rank N will have N real eigenvalues & all eigenvectors are orthogonal)
  - symmetric → $a_{ij} = a_{ji}$ & square

Eigenvalues tell us how a system behaves, evolves and transforms
↓
i.e. They provide deep insights into the matrix's properties and behavior
eg: → how a linear transformation will stretch or compress vectors
→ dynamical stability
→ observable quantities such as energy levels etc.

- Positive definite : A is positive definite if $\vec{x}^T A \vec{x} > 0$, $x \neq 0$
  ↳ All eigenvalues are positive

- Spectral radius : of a square matrix $\rho(A)$ is the largest absolute value of its eigenvalues
  ↳ $\rho(A) \leq \|A\|$

Methods of solving $Ax = b$

Direct methods

- Diagonal matrix
- Triangular "
  → forward / backward sub
- Cramer's rule $O(N!)$
- Gaussian elimination ($\frac{N^3}{3} + O(N^2)$ arithmetic operations)
  → Gauss - Jordan ($\frac{N^3}{2} + O(N^2)$)
  → LU decomposition
  → Thomas algorithm (for tridiagonal matrix)

banded = diagonally dominant.

require pivoting & equilibration (unless banded or positive definite)

Iterative Methods

- Jacobi iteration [$A = D + L' + U'$]
- Gauss - Seidel iteration [$A = D(L'' + I + U'')$]

convergence guaranteed if A is banded or positive definite

Error / Accuracy

All computations have round off errors → hence you get approximate solutions
↓
need to assess accuracy by comparing magnitude/size of different solutions
↓
Norms (works for scalars, vectors, tensors)

$$L_p(x) = \|x\|_p = \left(\sum_{i=1}^{N} |x_i|^p\right)^{1/p} \quad \text{(unweighted)}$$

$$L_p(x) = \|x\|_{p,w} = \left(\sum_{i=1}^{N} w_i |x_i|^p\right)^{1/p} \quad \text{(weighted)}$$

a = approx.
e = exact

$A_a = A_e + \delta A$

$x_a = x_e + \delta x$

$A_a x_a = b_e$

$$\text{Relative error} = \frac{\|\delta x\|}{\|x_e + \delta x\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A_e\|}$$

$\kappa(A)$: Spectral condition number $\left(= \frac{\lambda_{max}}{\lambda_{min}}\right)$
(measure of how sensitive the solution of a linear system $Ax = b$ is to

Spectral condition number $\left(= \frac{\lambda_{max}}{\lambda_{min}}\right)$

(measure of how sensitive the solution of a linear system $Ax=b$ is to small changes in b)

$\kappa(A) \approx 1$ (well-conditioned)

If $\kappa(A) >> 1$ (ill-conditioned)

↳ Iterative refinement: improving accuracy of the solution

- Calculate residual (using twice precision) $\vec{r}$
- Calculate error making this residual LU $\vec{\delta x} = \vec{r}$
- Subtract error from original solution $x_a = x_a - \delta x$
- Check $\left|\left| \frac{\delta x}{x_a} \right|\right| < \varepsilon$ [relative error]
- End loop

Partial Differential Equations

Phenomena where changes occur over multiple variables → very real-world applicable

$$F(x, y, \ldots, U_x, U_y, \ldots, U_{xx}, U_{yy}, U_{xy} \ldots) = 0$$

U_x ↳ $\frac{\partial U}{\partial x}$; U_{xx} ↳ $\frac{\partial U}{\partial x^2}$

↓ functional relationship among independent vars, dependent var & the partial derivatives

U → dependent var

x, y → independent vars

- Order → determined by highest-order PDE in the entire expression

1) Linear → F is a linear function of the dependent variable & its partial derivatives

$$x U_x + x_y U_y + \underbrace{U_{xy}}_{2^{nd}\ order} = 0$$

2) Almost linear → small or mild non-linearities

3) Quasi-linear → F is linear in highest order partial derivative, but non-linear in the dependent variable and/or lower-order derivatives

$$\underbrace{U U_x}_{\text{non-linearity}} + U_y = x_y$$

but U_x & U_y are linear (i.e. no U_x^2 or $\sin(U_x)$)

4) Non-linear → F is non-linear in any or all of the highest order partial derivatives appearing in the equation

$$U U_x + U_y^2 = x_y$$

E.g: ↳ $U^2 U_{xx} + U_y = x y^2$ → quasi-linear

↳ $U_x U_{xy} = \sin(xy)$ → Nonlinear

↳ This doesn't matter since U & its derivatives are what we check for

Solving:

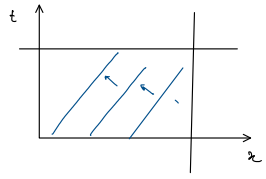
- First-order linear + quasi-linear

↳ Characteristic curves (by converting system of ODE's where family of curves in the $x_i - U$ space)

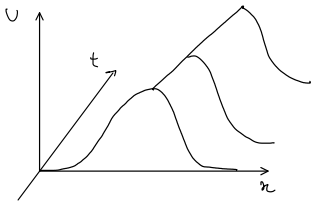
↓

along the curves, PDE is converted to ODE requiring one BC / IC

eg: $U_t + cU_x = 0 \rightarrow x = c_o + ct$







Weak vs. strong solutions

System of ODE's: $U_t + AU_x = 0$ (A: eigenvalues)
- → Elliptic
- → Hyperbolic
- → Mixed

• Second-order linear & quasi-linear PDE's

real or complex characteristic curves; may have two curves passing through each point

$$a_{ij} U_{x_i x_j} + b = 0$$

a_{ij} & b are functions only of x_i, U and U_x → quasi-linear

↳ Quasi-linear are further defined: $A = a_{ij}$

problems w/ evolutionary character:
- → Ultrahyperbolic (all eigenvalues are non-zero & at least two positive & two negative)
- → Hyperbolic (all eigenvalues are non-zero & all but one has same sign)
- → Parabolic (at least one eigenvalue is zero)

equilibrium problems:
- → Elliptic (All eigenvalues of A have same sign)
- → Mixed

02/24

Monday, 24 February 2025 9:30 AM

Midterm → March 7

↳ everything unit lecture 13

↳ Today's lecture.

FD vs. FV

FD
- point/node function
- ↓
- replace derivative/PDE w/ difference function

FV
- Control volume average solution
- ↓
- Has orders of accuracy O^1, O^2
- ↳ Figure/app. fluxes that go into the CV

Infinite no. of FV & FD for a given PDE

Method of lines

→ Space & time are decoupled
→ time difference first
→ & the space
- ↓ → backward / forward difference
- diffusion (central difference)

Convergence → as you refine the grid, solution goes to exact solution

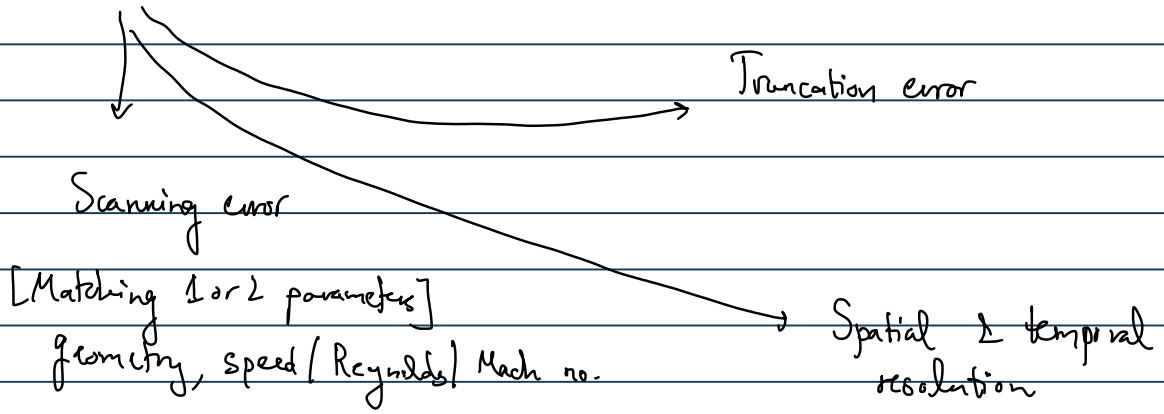
Lax equivalent theorem → linear, well-posed & consistent (unconditionally) & stable then it is convergent

Conservation of errors → guaranteed by FV method

Conservation of errors → guaranteed by FV method
↳ flux coming out from one cell must be the flux in of the next touching cell

Dissipation, diffusion errors.

Errors :



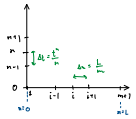
Monotonicity

Positivity

→ To ensure that the oscillations due to discontinuities don't go to crazy

$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ [Unsteady 1D, conduction]

1) Discretize time & space domain $T_i^n = T(x_i, t^n)$



Need m equations to solve for value of T at every grid point

Transforming PDE to FDEs (finite difference equation): We use the Taylor expansion

$$f(x+\Delta x, y) = f(x,y) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{\Delta x^2}{2!} + \frac{\partial^3 f}{\partial x^3}\frac{\Delta x^3}{3!} + \cdots$$

$$f(x-\Delta x, y) = f(x,y) - \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{\Delta x^2}{2!} - \frac{\partial^3 f}{\partial x^3}\frac{\Delta x^3}{3!} + \cdots$$

$$\therefore \underbrace{\frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}}_{\text{difference operator}} = \underbrace{\frac{\partial f}{\partial x}}_{\text{derivative}} + \frac{1}{2\Delta x}\underbrace{\left[\frac{2\Delta x^3}{3!}\frac{\partial^3 f}{\partial x^3} + \frac{2\Delta x^5}{5!}\frac{\partial^5 f}{\partial x^5} + \cdots\right]}_{\text{truncation error (TE)}}$$

$\mathcal{O}(\Delta x^2)$ → second order in mesh spacing

Thus the difference operator is second-order accurate

→ If f is single-valued, finite and cont. The taylor series for f is convergent
  ↳ This means that as $\Delta x \to \infty$, TE for higher order approx. will be smaller than that of a lower-order TE

→ Similarly, from the Taylor expansion we get:

"central difference" $\frac{f(x+\Delta x, y) - 2f(x,y) + f(x-\Delta x, y)}{\Delta x^2} \approx \frac{\partial^2 f}{\partial x^2}$; $TE = \mathcal{O}(\Delta x^2) = \frac{2\Delta x^2}{4!}\frac{\partial^4 f}{\partial x^4} + \frac{2\Delta x^4}{6!}$

Explicit Method: finding value for n+1, where solution @ n is assumed to be known ← This is the given initial cond. $T(x, t=0) = T_0$

↓
This is just because an algebraic expression

i.e. FDES which use no information at the new time level give uncoupled ODE's

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha\left(\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}\right)$$

Implicit Method → Coupled eq. where spatial information about the new time level is needed (but we only know solution at time level n)
↓
Therefore all eq. need to solved simultaneously → $A\vec{x} = \vec{b}$

We could also used a generalized approach of both explicit & implicit → This gives second-order accurate in both space & time

How based on the PDE → There are general requirements for BC's in order to get a solution

↳ hyperbolic, 1st order → need BC only at points where characteristic base curves enter the domain

↳ parabolic / elliptical → BC at every point on the boundary of spatial domain

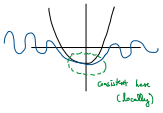
Consistency, Stability & convergence

- Discretization error → replacing PDES w/ FDES
- Stability error → round off errors (finite digits used in computations)
- Truncation error → $\mathcal{O}(\Delta t, \Delta x^2)$ error due to approximation

1) Consistency

$$\lim_{\substack{(\Delta x \to 0) \\ \Delta t \to 0}} (FDE - PDE)_i^n = \lim_{\Delta x \to 0} (TE)_i^n = 0$$

How well does the FDE locally approx the PDE



How to check?

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2}\right)$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha\left(\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}\right)$$

Expand all terms via Taylor series & then substitute:

$$T_{i+1}^n = T_i^n + \left(\frac{\partial T}{\partial x}\right)_n^i \Delta x + \cdots$$
$$T_{i-1}^n = T_i^n - \left(\frac{\partial T}{\partial x}\right)_n^i \Delta x + \cdots$$
$$T_i^{n+1} = T_i^n + \left(\frac{\partial T}{\partial t}\right)_i^n \Delta t + \cdots$$

↓
Let $\Delta t \to 0$, $\Delta x \to 0$ → if same as PDE, then consistent

↳ Unconditional vs Conditional consistency
↓ no dependence on rate of Δx & Δt ↳ depends on relative rates of Δx & $\Delta t \to 0$

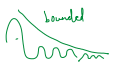
↳ Since Δx & Δt are finite computationally → consistency alone does not guarantee accuracy
there a consistent solve still will have drift

no dependence on ratio of Δx & Δt

depends on relative ratio of Δx & $\Delta t \to 0$

↳ Since Δx & Δt are finite computationally → consistency alone does not guarantee accuracy.
Thus a consistent soln. still behaves diff. than PDE

2) (Numerical) stability :
+ when time domain is involved (i.e. accumulation or growth of errors) (parabolic & hyperbolic PDE's)
+ Depends solely on FDE ; can be bounded (i.e. don't grow)
+ Computational domain must include all of the analytical domain



$\phi = \frac{\alpha \Delta t}{\Delta x^2} \le \frac{1}{2}$ → conditionally stable
↑ dimensionless parameter

Any chosen Δt, Δx → unconditionally stable (most implicit methods)

How to check ?
- Discrete perturbation → perturbation ε @ i & n
- Fourier method → disturbance ε at every grid point at an arbitrary time n
- Matrix method
- Modified equation method

$\left|\frac{\Delta T}{\varepsilon}\right| \le 1$ → perturbations cannot grow
& $\frac{\Delta T}{\varepsilon} > 0$ decay must be monotonic
} Hence need to do for only one time step

- Obtain an evaluation eq: i.e. w/ ε as $f(\Delta x, \Delta t)$
- Use fourier series expansion : $\varepsilon_i^n = \sum_{-\infty}^{\infty} A_j^n e^{Ijhx}$ (fourier component (Amplitude A_j^n, wave. no. = h_j); amplification factor)
- Analyze error in the components $A_j^{n+1} = G_j A_j^n$
- Accounts well for periodic BC (but other BC's may be an issue)

Assume distribution of errors along the x-axis is given by a fourier series in x & time-wise variation is exponential in t
↓
$\varepsilon(x,t) = e^{\alpha t} \sum_m e^{Ikx}$

$G = \left|\frac{\varepsilon_i^{n+1}}{\varepsilon_i^n}\right| \le 1$ [Errors should not be growing]

↳ For $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$
↓
$\frac{\Delta t}{(\Delta x)^2} \le \frac{1}{2}$ [stability requirement]

$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$
$C = c\frac{\Delta t}{\Delta x} \le 1$
↓ courant number
(from von-Neumann stability analysis) [called CFL condition]

FDE's at every interior & boundary grid point are written in matrix form
$T^{n+1} = AT^n + b$
- introduce disturbance to get $\varepsilon^{n+1} = A^{(n+1)}\varepsilon^0$

If A is convergent matrix, e^{n+1} will approach zero in the limit as $n \to \infty$
↓
$\|A\| \le 1$; $\rho(A) \le 1$
↑ smallest norm = spectral radius

3) Convergence : If total error (discretization + stability) → 0 as $\Delta x \to 0$, $\Delta t \to 0$
+ If FDE is convergent, then numerical soln of any accuracy can be obtained by changing the grid-spacing & time-step.
+ FDE must give same solution as PDE in some limiting cases

- Lax-Wendroff method to derive an FDE
  ↳ express time derivative in terms of spatial derivatives
  ↳ Use Taylor series for each point
- Courant no. [$C_o = \frac{c\Delta t}{\Delta x}$]

If $C_o = 1$, then truncation error is zero

For example :

$u_t = K u_{xx}$; $K > 0$; $u(t,x)$

$\frac{(u_i^{n+1} - u_i^n)}{\Delta t}$ ↓ can also be written as

$= \frac{u_i^n(t+\Delta t, x) - u_i^n(t,x)}{\Delta t}$

$= \frac{\left[u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2\right] - u_i^n}{\Delta t}$

Taylor series:
$f(x)\Big|_{x=a} = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 \cdots$
$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$

$\therefore \quad u_t = \left(\frac{\partial u}{\partial t}\right)_i^n + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t}{2} + O(\Delta t^2)$

Similarly $K u_{xx} = \frac{1}{\Delta x^2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)$
($u(t, x+\Delta x)$, $u(t, x-\Delta x)$)

$= \frac{1}{\Delta x^2}\left[\left(u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2} + \cdots\right) - 2u_i^n + \left(u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2}\right)\right]$

$= \frac{1}{\Delta x^2}\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^4}{4!} \cdot 2 + \cdots\right]$

$= \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^2}{12} + O(\Delta x^2)$

$\therefore$ Modified equation : $u_t - K u_{xx} = 0$

$\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{\Delta t}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n + O(\Delta t^2) = K\left[u_{xx} + \frac{\Delta x^2}{12} u_{xxxx} + O(\Delta x^4)\right]$

Neglecting higher order terms:

$\frac{\partial u}{\partial t} - K\frac{\partial^2 u}{\partial x^2} = K\frac{\Delta x^2}{12}\frac{\partial^4 u}{\partial x^4} - \underbrace{\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2}}$

this makes it hyperbolic

tt, t, xx, xxxx

Error Analysis

Hard to show convergence → need FDE's to model physical properties/parameters

Conservation principle → In - out + gen = Acc. for a CV
"conservative property"

For fluid flow, a disturbance at any location in the flows field can spread to other locations by three transport mechanisms

- Convection (due to fluid motion)
- Diffusion
- Pressure waves

} FDE's have transportive property

spreads in every direction by random molecular motion → but disturbance is smoothed out by process called dissipation

If $\frac{\partial}{\partial t}$ (1st order in time), then even-order spatial derivatives are dissipation, i.e. $\mu \nabla^2 u$, $\mu \nabla^2 v$

truncation error w/ leading even order space derivatives are termed as dissipative terms

even likes to "diss"

Given a PDE, we can define multiple FDE's
↳ Now of the FDE's we can write modified equations (Taylor expansion)
↳ The order of dissipation of an FDE is the lowest even-order spatial derivative of the modified equation
↳ For ex: Pg 65, book
↳ A 4th order dissipation means we will be less dissipation error than that of a first order

Dispersion → solutions (= waves w/ Fourier Transform) that have different wavelengths travel at different speeds } speed ∝ f(wavelength)
↓
error if waves not travelling at same speed.
All fluids are modelled as non-dispersive

↳ presence of odd-order space-derivative in the TE is responsible for dispersion

When there is $\frac{\partial}{\partial t}$ term (first order) → even derivatives = diffusion error (dissipation)
odd derivatives = dispersion error

↓
∴ coefficients of FDE should be same as PDE for even
to model property
↓
If not then dissipative error

03/05

Wednesday, 5 March 2025 9:32 AM

Midterm 03/07 !

→ Open book ; open notes ; 10 questions ; very picky (highlight all points)
(No internet)

→ Conceptual questions

Finite Volume Method

FD → b
↳ generalized coordinate
(mapping from cartesian to via jacobian

Euler equation : $\frac{\partial U}{\partial t} + \nabla \cdot F = 0$;

$F = fi + gj$

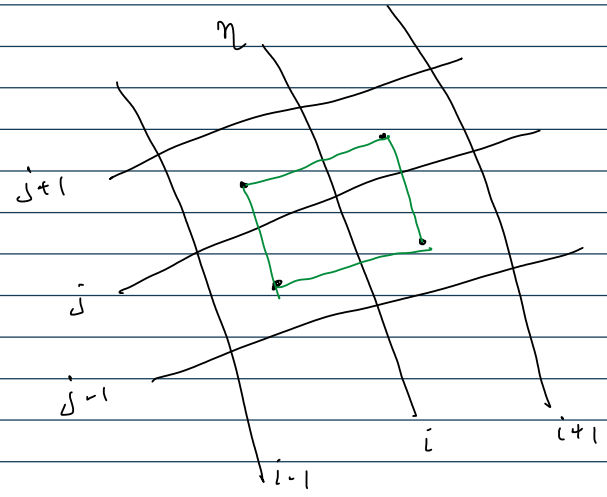
→ Still need to generate grid but do not have to map to generalized coordinate

→ Integrate over CV and use Gauss divergence theorem $\int_V (\nabla \cdot F)\, dV = \int_A F \cdot dA$

"What goes in, comes out"

$$V \frac{dU}{dt} + \int_A F \cdot dA = 0$$

→ If CV is moving → need Leibniz's rule



Flux . surface area

→ then choose discretized volume method

↓

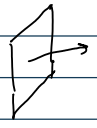
Backward, backward bias

Cell central method vs. vertex centered

How to calculate area of each face of a 3D cell ?

vectors on surface





$\vec{S}_1 = \frac{1}{2}(r_1 \times r_2)$

cell volume = $\frac{1}{3} r \left(S_1 + S_2 + S_2 \right)$

Converting FDM to FVM

1) Transform GE in generalized coord in strong conservation-law form

2) :

3) discretize PDE

Potential flow problem:

Potential flow is used to model irrotational where the viscosity effects are assumed to be negligible - typically externally flow over solid surfaces or flow at high Reynolds number. In such cases, the thickness of the boundary layer is assumed to be much smaller than the dimensions of the object.

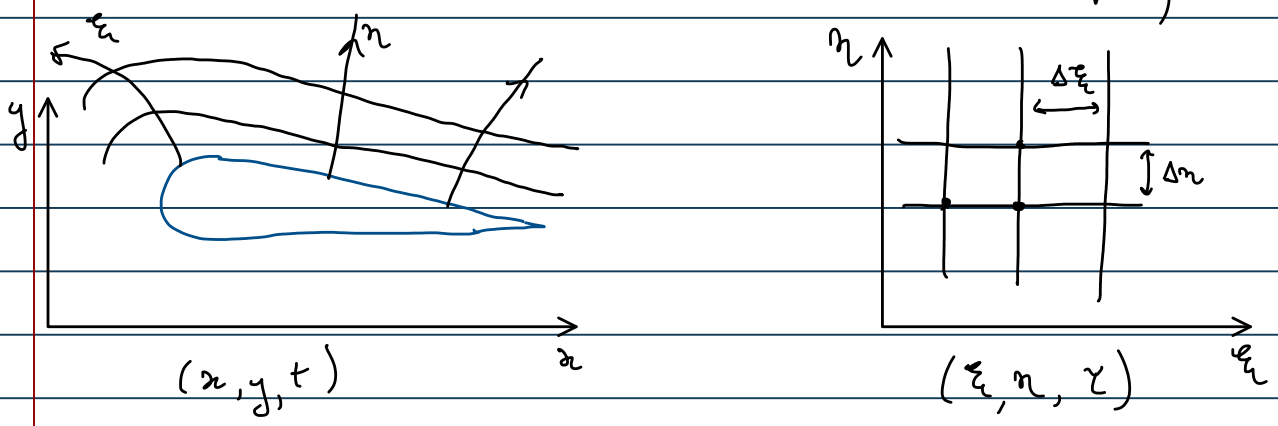
The given equation is the Laplace equation using stream function. Stream function (ψ) can be defined for any 2D flow such that the stream function is constant along a streamline. Additionally, change in stream function between a pair of streamlines equals the volummetric flowrate between the two streamlines. Finally, lines of constant streamlines are perpendicular to the lines of constant velocity potential (Φ).

Review 4.0

Saturday, 22 March 2025 7:14 PM

Grid generation

Physical grid ⟶ Computational grid (Uniform)



$$\xi = \xi(x, y, t)$$

$$\eta = \eta(x, y, t)$$

$$\tau = \tau(t)$$

$$\text{Jacobian matrix } (J) = \frac{\partial(x,y)}{\partial(\xi,\eta)} = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{vmatrix}$$

What does conservative form mean?

$$\frac{\partial}{\partial t}\int \rho \vec{v}\, dV$$

↳ There is no derivative outside i.e. nothing outside derivative here: $\frac{\partial}{\partial t}(m) + \frac{\partial}{\partial x}$ } No derivative outside the outermost derivative

$$(x, y, t) \Longleftrightarrow (\xi, \eta, \tau)$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial t}$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \tau}\frac{\partial \tau}{\partial x} + \cdots$$

$$\frac{\partial^2 u}{\partial y^2} = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right)$$

$$\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)$$

} This is usually sufficient to express in terms of chain rule

↓

strong conservative ($\times J$)

$$J\xi_y \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial \xi}\left[J\xi_y \frac{\partial u}{\partial y}\right] - \frac{\partial u}{\partial y}\frac{\partial}{\partial \xi}(J\xi_y) \quad \checkmark$$

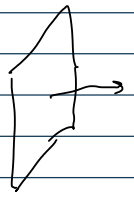
↳ How to write geometric conservation laws

↓

There are 4 general ones → $\frac{\partial}{\partial \xi}\xi_x J + \frac{\partial}{\partial \eta}(\eta_x J) = 0$

If grid is not deformed → $\frac{\partial J}{\partial \tau} = 0$ & $\frac{\partial}{\partial t} = \frac{\partial}{\partial \tau}$

↳ Metric coefficients → (has a physical meaning)



$\vec{A}_\xi = J \nabla \xi$

$= J\xi_x \vec{i} + J\xi_y \vec{j} + J\xi_z \vec{k}$

$\xi_x J = A_x$

$V_{i,j,k} = J_{i,j} \Delta\xi \Delta\eta \Delta\zeta$

Metric coefficients × Jacobian represent the area in the physical domain of the faces of the cell ★

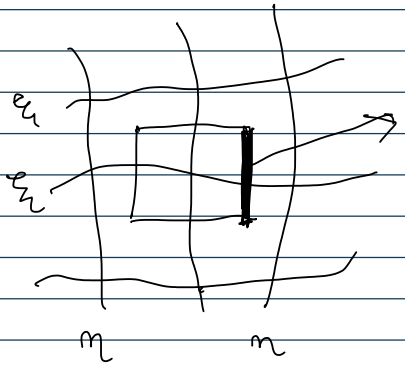
→ This is how you should calculate stuff .... i guess...

(Q2)

Multiplied by J^2 $\quad \frac{\partial^2 \tau}{\partial x^2} + \cdots$

$$\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)\tau \quad \cdots$$

$$J^2 \xi_x \frac{\partial}{\partial \xi}\left(\xi_x \frac{\partial \tau}{\partial \xi}\right)$$



$\vec{A_S} = (J\xi_x)\vec{i} + J(\xi_y)\vec{j}$

$\vec{A_S} = A_x \vec{i} + A_y \vec{j}$

03/31

Monday, 31 March 2025 9:31 AM

Strong conservation → no source terms created by the mapping / transformation

There is also weak conservative →

Method of lines → discretize space & time which are decoupled

Review 5.0

Tuesday, 8 April 2025 9:12 AM

Conservative & Non-conservative

Equations obtained from directly applying fundamental physical principles to a finite control volume

↓

i.e. eq. in integral form (that are then applied to obtain PDE's)

Entire system of governing equations in conservation form } continuity, momentum & energy

unsteady (solution vector) | flux terms | source term

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = S$$

"Shock" → sharp, discontinuous changes in the primitive flow-field variables (p, ρ, u, T)

↳ Conservation form of equations should be used to get smooth + stable (or else you get wiggles

↳ because conservation form in fact uses (ρu) & product of dependent variables ↳ which are generally conserved in a shock!

(i.e. changes in flux are either ~~zero~~ or small across a shock wave)

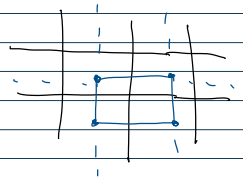
Office hours:

+ Two-boundary method for grid generation

↓

Jacobian → Area in 2D

↳ Volume in 3D



↳ Local Thermodynamic equilibrium

↓

When can you assume it?

If it is changing

→ Equations of state ($PV = nRT$)

→ connect measurable/non-measurable variables

↓ ↳ energy (

T, ρ

↳ Gaussian elimination (Accuracy)

↳ Condition no.

↳ residual

↳ Iterative refinement →

↳ Modified equation & phase angles

↳ Modified equation & phase angles

ϕ - phase angle relates to odd derivatives terms

↳ Implicit vs. Explicit ✓

↳ FDE's

↳ Errors

Types of PDE's & their significance:

$u = u(x,y)$

$v = v(x,y)$

$$\begin{matrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial x} \end{matrix}$$

Making matrix & checking determinant/ eigenvalues

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

characteristic lines

↓

derivative of u & v are indeterminant

(i.e. derivative of solution do not have a unique solution)

↓

PDE's simplify to ODE's

→ Gives type of PDE's

For first order: hyperbolic, parabolic, elliptic (quasilinear)

- information at a given point P influences only those regions b/w advancing characteristics
- information at point P in the xy-plane influences the entire region of the plane to one side of P
- info at P influences all other regions of the domain → entire boundary must be defined.

eg:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} = 0$$

04/14

Monday, 14 April 2025 9:44 AM

Flux-differencing has dissipative errors

↳ Godunov scheme

↳ Reimann problem } take a solution & average w/ time → Then average again

↓

Solve the reimann problem exactly

↳ App. Reimann solver schemes

↳ Osher's method

↳ Roe's method

↓

Mean value theorem

Concept: → solving app. reimann sum where a is a constant

↓

and we use mean value theorem

Reimann solvers are always 1st order accurate

04/21

Monday, 21 April 2025 9:34 AM

Exam 2 → May 5th 1-3pm [open book, open notes] 12 questions, 5 points
One problem all throughout

HW7 → due May 27th

Upwind difference → need to know HW7

→ Explicit methods biggest problem

time step for stability is very small (stability criteria very stringent & it gets worse)

→ Implicit methods biggest issue

↳ need to have non-linearization
[time linearization → newton-rhapson]

↳ Need a bigger stencil (higher matrix bandwidth) → Need LU methods or ADI (factorization)

↳ How to address time steps

Time-split method

↳ Unconditionally stable for linear [but conditionally stable for non-linear]

Accelerating Convergence Rate

Explicit methods: (How to increase computational speed)
+ residual smoothing
+ local time stepping
+ Multigrid

error is removed quickly (neighbors seen right away) } h then 2h & then 4h

Compressible flow of a non-reacting ideal gas

BC's: Inflow / outflow → use euler equation BC's (momentum dominates over viscous, reynolds no. is high, diffusion in streamwise direction is negligible)

↓

else have to use full Navier stokes

exceptions: high Re, but if grid gets really small then need to have full info

04/23

Wednesday, 23 April 2025 9:33 AM

→ Velocity BC on wall

→ Velocity BC on inviscid wall

→ Pressure & T

Inflow :

Incompressible flow → harder to model since no time derivative
(speed of sound is infinite → any change is
affected everywher

Last class

Sunday, April 27, 2025 2:37 PM

Boundary conditions

↳ Dirichlet boundary conditions

Prime factorization

↓

Problem 1 from HW7

Flux vector splitting:

↳ spectral radius

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

$$c = c^+ + c^-$$

Local eigenvalue

$$c^+ = \frac{1}{2}(c + |c|)$$

$$c^- = \frac{1}{2}(c - |c|)$$

↓

less dissipating

→ First order upwind →

Central differencing → fourth order diffusion (& coefficient has to be negative for stability)

Given PDE:

1) Δu^{n+1} everywhere Δ onset

2) Linearize

$$D_x \cdot (u^{n+1})^2 = (u^n)^2 + 2u^n \Delta u^{n+1}$$

$$\underbrace{D_x D_y (u^{n+1}(x,y))}$$

$$u^{n+1} = u^n + \Delta u^{n+1} \approx u^n + \Delta u^n$$

$$\frac{\partial}{\partial x}\left(\alpha \frac{\partial \phi}{\partial x}\right) = \frac{\left(a \frac{\partial \phi}{\partial x}\right)_{i+\frac{1}{2}} - \left(a \frac{\partial \phi}{\partial x}\right)_{i-\frac{1}{2}}}{\Delta x}$$

$$= \frac{a_{i+\frac{1}{2}}\left(\frac{\phi_{i+1} - \phi_i}{\Delta x}\right) - a_{i-\frac{1}{2}}\left(\frac{\phi_i - \phi_{i-1}}{\Delta x}\right)}{\Delta x}$$

Boundary conditions for tridiagonal matrix

$$L_{ij} \Delta u^*_{i+1,j} + M_{i,j} \Delta u^*_{i,j}$$





→ Dirichlet → does not change w/ time

$T = 40\,C$

$\Delta u^{n}_{i+1,j} = 0$ (does not change)

→ Neumann → $\frac{\partial u}{\partial x} = q$

$\frac{u_2 - u_1}{\Delta x} = q \rightarrow u_1 = u_2 - q\Delta x$

$M \rightarrow \frac{V}{C}$

$T \rightarrow C = \sqrt{\gamma R T}$

V

T

$\frac{\partial^2 P}{\partial x^2} = 0$

$$\frac{P_2 - 2P_1 + P_0}{\Delta x^2} = 0 \rightarrow P_0 = 2P_1 - P_2$$

HW

Wednesday, 5 February 2025 9:26 AM

Problem 1

Heun method (2nd order Runge-Kutta)

$$2\frac{d^2y}{dx^2} + 3x^2y\frac{dy}{dx} = 10x^2$$

@ $x=1$, $y=2$, $\frac{dy}{dx}=2$

$$\frac{dy}{dx} = f(y,x)\ ;$$

$$\therefore 2\frac{df}{dx} + 3x^2yf = 10x^2$$

$$\frac{df}{dx} = 5x^2 - \frac{3}{2}x^2yf = g(x,y)$$

@ $x_0=1$, $f(1,2)=2$, $y=2$, $g(1,2)=-1$

Discretize the domain : $x^n = x_0 + n\Delta x$ (1, 0.1)

Discretize the ODE : $\frac{\tilde{y}^{n+1} - y^n}{\Delta x} = f(y^n, x^n) \rightarrow \tilde{y}^{n+1} = y^n + \Delta x\, f(y^n, x^n)$

$$\frac{f^{n+1} - f^n}{\Delta x} = \frac{f(y^n,x^n) + g(\tilde{y}^{n+1}, x^{n+1})}{2}$$

$$f^{n+1} = f^n + \frac{\Delta x}{2} g(\tilde{y}^{n+1}, x^{n+1})$$

$$f(y^{n+1}, x^{n+1}) = f(y^n, x^n) + \frac{\Delta x}{2}\left[5(x^{n+1})^2 - \frac{3}{2}(x^{n+1})^2 \tilde{y}^{n+1} f(\tilde{y}^{n+1}, x^{n+1})\right]$$

$f^{n+1} = f(y^{n+1}, x^{n+1})$; $f(x_0, y_0) = 2$; $g(x_0, y_0) = -1$

n	$x^n = x_0 + n\Delta x$	$\tilde{y}^n = y^{n-1} + \Delta x f^{n-1}$	$\tilde{f}^n = f^{n-1} + \Delta x\, g(\tilde{y}^n, x^n)$	$f^n = f^{n-1} + \frac{\Delta x}{2} g(\tilde{y}^n, x^n)$	$y^n = y^{n-1} + \frac{\Delta x}{2}(f^{n-1} + \tilde{f}^n)$
0	$x^0 = x_0 = 1$	—	—	—	—
1	$x^1 = 1 + (0.1) = 1.1$	$\tilde{y}^1 = y^0 + \Delta x f^0 = 2 + (0.1)(2) = 2.2$	$\tilde{f}^1 = f^0 + \Delta x\, g(\tilde{y}^1, x^1) = 2 + (0.1)(-1) = 1.9$	$f^1 = f^0 + \frac{\Delta x}{2} g(\tilde{y}^1, x^1) = 2 + \frac{(0.1)}{2}\left(5(1.1)^2 - \frac{3(1.1)^2(2.2)}{2}(1.9)\right) = 1.8732$	$y^1 = y^0 + \frac{(0.1)}{2}(2 + 1.9) = 2.195$
2	$x^2 = 1 + 2(0.1) = 1.2$	$\tilde{y}^2 = y^1 + \Delta x f^1 = 2.195 + (0.1)(1.8732) = 2.38823$	$\tilde{f}^2 = f^1 + \Delta x\, g(\tilde{y}^2, x^2) = 1.8732 + (0.1)(-1.4183) = 1.7732$	$f^2 = f^1 + \frac{\Delta x}{2} g(\tilde{y}^2, x^2) = 1.8732 + \frac{(0.1)}{2}\left(5(1.2)^2 - \frac{3(1.2)^2}{2}(2.38823)(1.732)\right) = 1.7072$	$y^2 = y^1 + \frac{(0.1)}{2}(f^1 + \tilde{f}^2) = 2.195 + \frac{(0.1)}{2}(1.8732 + 1.732) = 2.3$

$5(1.1)^2 - \frac{3}{2}(1.1)^2(2.195)(1.8732) = -1.4183$

Therefore ;

for $\Delta x = 0.1$ (w/ the given IC's)

we get

$y^1 = 2.195$, $f^1 = \frac{dy^1}{dx} = 1.8732$	@ $x = 1.1$
$y^2 = 2.3952$, $f^2 = \frac{dy^2}{dx} = 1.7172$	@ $x = 1.2$

Problem 2

2nd order Adams-Moulton method

$$2\frac{dy}{dx} - 3y^2 = 10 \quad , \quad y=10 \text{ at } x=1$$

$$\frac{dy}{dx} = \frac{5 - 3y^2}{2} = f(y,x) \ ; \quad y_0(x_0=1) = 10$$

$$f(y_0, x_0) = \frac{5 - 3(10)^2}{2} = -145$$

Implicit Adams-Moulton (Trapezoidal Method) :

$$y_{n+1} = y_n + \frac{\Delta x}{2}\left[f(y_n, x_n) + f(y_{n+1}, x_{n+1})\right]$$

@ $n=0$, $y_1 = y_0 + \frac{\Delta x}{2}\left[f(y_0,x_0) + f(y_1,x_1)\right]$

$$y_1 = 10 + \frac{(0.1)}{2}\left[-145 + 5 - 3y_1^2/2\right]$$

$$y_1 = 10 - 7 - 0.075\, y_1^2$$

$$0.075\, y_1^2 + y_1 - 3 = 0$$

Solving using Newton Rhapson : Let $h = 0.075\, y_1^2 + y_1 - 3$

$$h' = 0.15\, y_1 + 1$$

$$y_1^{(k+1)} = y_1^{(k)} - \frac{h(y_1^{(k)})}{h'(y_1^{(k)})}$$

@ $k=0$,

$$y_1^{(1)} = y_1^{(0)} - \frac{h(y_1^{(0)})}{h'(y_1^{(0)})}$$

estimate using Euler → $y_1^{(0)} = y_0 + \Delta x\, f(y_0) = 10 + (0.1)(-145) = -4.5$

$h(y_1^{(0)}) = h(-4.5) = 0.075(-4.5)^2 + (-4.5) = -5.921875$

$h'(-4.5) = 0.15(-4.5) + 1 = 0.325$

2 BVP → Problem 3 → L & T

$\therefore\ y_1^{(1)} = 13.9$

@ k=1,

$h(13.9) = 0.075(13.9)^2 + 13.9 - 3 = 25.39075$

$h'(13.9) = 0.15(13.9) + 1 = 3.085$

$y_1^{(2)} = 13 - \frac{25.39075}{3.085} = 5.67 \rightarrow y_1 \sim 5.67$

Using quadratic formula

$y_1 = \{-15.86,\ 2.5\}$ ✓ → still far off w/ two iterations but will just use this value

@ n=1 ; $y_2 = y_1 + \frac{\Delta x}{2}\left[f(y_1, x_1) + f(y_2, x_2)\right]$

$f(y_1, x_1) = 5 - \frac{3}{2}y_1^2 = 5 - \frac{3}{2}(5.67)^2 = -43.22$

$y_2 = y_1 + \frac{\Delta x}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$ (5.67)

$y_2 = 5.67 + \frac{(0.1)}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y = 3.759 + 0.075y_2^2$

$0.075y_2^2 + y - 3.759 = 0$

Solving using Newton Rhapson : Let $g = 0.075y_2^2 + y_2 - 3.759$

$g' = 0.15y_2 + 1$

$y_2^{(k+1)} = y_2^{(k)} - \frac{g(y_2^{(k)})}{g'(y_2^{(k)})}$

@ k=0,

$y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$

estimate using Euler → $y_2^{(0)} = y_1 + \Delta x f(y_1) = 5.67 + (0.1)(-43.22) = 1.348$

$g(y_2^{(0)}) = g(1.348) = 0.075(1.348)^2 + (-2.411) = -2.2747$

$g'(1.348) = 0.15(1.348) + 1 = 1.2022$

$\therefore\ y_2^{(1)} = 1.348 - \frac{(-2.2747)}{(1.2022)} = 3.24$

@ k=1,

$g(13.9) = 0.075(3.24)^2 + 3.24 - 3.759 = 0.26832$

$g'(13.9) = 0.15(3.24) + 1 = 1.486$

$y_2^{(2)} = 3.24 - \frac{0.26832}{1.486} = 3.0594 \rightarrow y_2 \sim 3.06$

Using quadratic formula

$y_2 = \{-16.39,\ 3.0587\}$ ✓ → much better approximation

$\therefore$ For

$x_1 = 1.1,$	$y_1 = 5.67$	w/ $\Delta x = 0.1$
$x_2 = 1.2,$	$y_2 = 3.06$	& given initial conditions

Problem 3

Steady state conduction in a flat plate

L ⇒ plate thickness

$\dot{q}$ ⇒ heat generation

$\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q} = 0$

$k = a + bT$

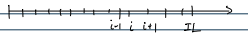
$\dot{q} = d + eT$

$-k\frac{dT}{dx}\Big|_{x=0} = h_H\left(T_H - T(x=0)\right)$

$-k\frac{dT}{dx}\Big|_{x=L} = h_L\left(T(x=L) - T_c\right)$

$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$ } $F_1(T_1, T_2)$ ($k_1 = a + bT_1$)

$-k_1\frac{(T_{IL} - T_I)}{\Delta x} = h_L(T_{IL} - T_c)$ } $F_{IL}(T_{IL}, \ldots)$



Discretizing the equation :

$\left[\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q}\right]_i = \frac{k\frac{dT}{dx}\Big|_{i+\frac{1}{2}} - k\frac{dT}{dx}\Big|_{i-\frac{1}{2}}}{\Delta x} + q_i = 0$

$= \frac{\left(k_{i+\frac{1}{2}}\frac{T_{i+1} - T_i}{\Delta x}\right) - \left(k_{i-\frac{1}{2}}\right)\frac{T_i - T_{i-1}}{\Delta x}}{\Delta x} + d + eT_i = 0$ } $F_i(T_{i+1}, T_i, T_{i-1}) = 0$, $i = 2, 3, \ldots, IL-1$

$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2}$

$$= \frac{k_{i+\frac{1}{2}}\left(\frac{\Delta x}{2}\right)^2 + k_{i-\frac{1}{2}}\left(\frac{\Delta x}{2}\right)^2}{\Delta x} + d + eT_i = 0$$

$$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2}$$

$$k_{i-\frac{1}{2}} = \frac{k_i + k_{i-1}}{2} = \frac{(a+bT_i) + (a+bT_{i-1})}{2}$$

$$F_i(T_{i+1}, T_i, T_{i-1}) = 0, \quad i = 2, 3, \ldots, IL-1$$

Translating this in the form of $Ax = b$:

$$\frac{(k_{i-1/2})}{(\Delta x)^2} T_{i-1} - \left[\frac{(k_{i+1/2})}{(\Delta x)^2} + \frac{(k_{i-1/2})}{(\Delta x)^2} - e\right] T_i + \frac{(k_{i-1/2})}{(\Delta x)^2} T_{i-1} = -d$$

Need the Newton Raphson (product of T_{ij}) → find derivatives

@ boundaries:

F_1 & F_{IL} apply

∴ Matrix A will be constructed as:

$$\begin{bmatrix} A_{i,i+1} = \frac{k_{i+1/2}}{(\Delta x)^2} & & \\ & A_{i,i} = -\left(\frac{(k_{i+1/2}) + (k_{i-1/2})}{(\Delta x)^2} + e\right) & \\ & & A_{i,i-1} = \frac{(k_{i-1/2})}{(\Delta x)^2} \end{bmatrix} \begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} BC\#1 \\ -q_i \\ BC\#2 \end{bmatrix}$$

Boundary conditions

$$-k\left.\frac{dT}{dx}\right|_{x=0} = h_H(T_H - T(x=0))$$

$$-k\left.\frac{dT}{dx}\right|_{x=L} = h_L(T(x=L) - T_c)$$

$$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$$

$$-k_1\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$$

$$\frac{-k_1}{\Delta x}T_2 + \frac{k_1}{\Delta x}T_1 = h_H T_H - h_H T_1$$

$$\left(\frac{k_1}{\Delta x} + h_H\right)T_1 - \frac{k_1}{\Delta x}T_2 = h_H T_H$$

$A[0,0]$ $A[0,1]$ boundary $[0]$

$$\frac{-k_1}{\Delta x}T_{IL-1} + \frac{k_1}{\Delta x}T_{IL} = h_L(T_{IL} - T_c)$$

$$\left(\frac{-k_1}{\Delta x} - h_L\right)T_{IL} + \left(\frac{k_1}{\Delta x}\right)T_{IL-1} = -h_L T_c$$

$A[-1,-1]$ $A[-1,-2]$

$$F(T_{i-1}, T_i, T_{i+1}) = (a+bT$$

Newton Raphson

$$\begin{bmatrix} \frac{\partial F_1}{\partial T_1} & \frac{\partial F_2}{\partial T_2} \\ \cdots & \cdots \end{bmatrix}$$

The above algorithm is coded in python (w/ the help of Chatgpt to debug the code) and the following plots are obtained

The decrease temperature (from hot end of ~2000 K) to cold end of ~ 400 K makes sense since the rod is cooling down.

Additionally, the grid spaces were varied from 100 to 10000 to give the same results (grid independent)

Thomas

Office hrs:

$u(x,y) = 0$ } Need to solve for this → start w/ a guess

$v(x,y) = 0$

$$\begin{bmatrix} \frac{\partial u}{\partial x}\Big|_{(x_i, y_i)} & \frac{\partial u}{\partial y}\Big|_{(x_i, y_i)} \\ \frac{\partial v}{\partial x}\Big|_{x_i, y_i} & \frac{\partial v}{\partial y}\Big|_{x_i, y_i} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = - \begin{bmatrix} u(x_i, y_i) \\ v(x_i, y_i) \end{bmatrix}$$

Constants

Constants

Thomas

$Ax = b$ (Thomas algo.)

↳ Update guess next

$$x^{i+1} = x^i + \Delta x$$
$$y^{i+1}$$

↳ Iterate until

$$\left| (\text{norm}\left[\Delta x^{i+1}, y^{i+1}\right]) - \text{norm}\left[\Delta x^i, y^i\right] \right| < \varepsilon = 0$$

↳ Incorporate the boundary conditions

Problem 1

Heun method (2nd order Runge-Kutta)

$$2\frac{d^2y}{dx^2} + 3x^2y\frac{dy}{dx} = 10x^2$$

@ $x=1$, $y=2$, $\frac{dy}{dx}=2$

$$\frac{dy}{dx} = f(y,x)\ ;$$

$$\therefore\ 2\frac{df}{dx} + 3x^2yf = 10x^2$$

$$\frac{df}{dx} = 5x^2 - \frac{3}{2}x^2yf = g(x,y)$$

@ $x_0=1$, $f(1,2)=2$ ($y=2$), $g(1,2)=-1$

Discretize the domain: $x^n = x_0 + n\Delta x$, $\Delta x = 0.1$

Discretize the ODE: $\frac{\tilde{y}^{n+1}-y^n}{\Delta x} = f(y^n,x^n) \rightarrow \tilde{y}^{n+1} = y^n + \Delta x\, f(y^n,x^n)$

$$\frac{f^{n+1}-f^n}{\Delta x} = \frac{f(y^n,x^n)+g(\tilde{y}^{n+1},x^{n+1})}{2}$$

$$f^{n+1} = f^n + \frac{\Delta x}{2}g(\tilde{y}^{n+1},x^{n+1})$$

$$f(y^{n+1},x^{n+1}) = f(y^n,x^n) + \frac{\Delta x}{2}\left[5(x^{n+1})^2 - \frac{3}{2}(x^{n+1})^2\tilde{y}^{n+1}f(\tilde{y}^{n+1},x^n)\right]$$

$f^{n-1} = f(y^{n-1},x^{n-1})$; $f(x_0,y_0)=2$

g ; $g(x_0,y_0)=-1$

n	$x^n = x_0 + n\Delta x$	$\tilde{y}^n = y^{n-1} + \Delta x f^{n-1}$	$\tilde{f}^n = f^{n-1} + \Delta x\, g(\tilde{y}^{n-1},x^{n-1})$
0	$x^0 = x_0 = 1$	—	—
1	$x^1 = 1+(0.1) = 1.1$	$\tilde{y}^1 = y^0 + \Delta x f^0 = 2+(0.1)(2) = 2.2$	$\tilde{f}^1 = f^0 + \Delta x\, g(\tilde{y}^0,x^0) = 2+(0.1)(-1) = 1.9$
2	$x^2 = 1+2(0.1) = 1.2$	$\tilde{y}^2 = y^1 + \Delta x f^1 = 2.195 + (0.1)(1.8732) = 2.3823$	$\tilde{f}^2 = f^1 + \Delta x\, g(\tilde{y}^1,x^1) = 1.8732 + (0.1)(-1.4122) = 1.732$

$5(1.1)^2 - \frac{3}{2}(1.1)^3(2.195)(1.8732) = -1.4122$

$f^n = f^{n-1} + \frac{\Delta x}{2}g(\tilde{y}^n,x^n)$	$y^n = y^{n-1} + \frac{\Delta x}{2}(f^{n-1}+\tilde{f}^n)$
—	—
$f^1 = f^0 + \frac{\Delta x}{2}g(\tilde{y}^1,x^1) = 2 + \frac{(0.1)}{2}\left(\underbrace{5(1.1)^2 - \frac{3(1.1)^2(2.2)(1.9)}{2}}_{5(1.1)^2-\frac{3}{2}(1.1)^2\tilde{y}^1\tilde{f}^1}\right) = 1.8732$	$y^1 = y^0 + \frac{(0.1)}{2}(2+1.9) = 2.195$
$f^2 = f^1 + \frac{\Delta x}{2}g(\tilde{y}^2,x^2) = 1.8732 + \frac{(0.1)}{2}\left(5(1.2)^2 - \frac{3(1.2)^2(2.3823)(1.73)}{2}\right) = -1.707$	$y^2 = y^1 + \frac{(0.1)}{2}(f^1+\tilde{f}^2) = 2.195 + \frac{(0.1)}{2}(1.8732+1.732) = 2.3752$

Therefore; for $\Delta x = 0.1$ (w/ the given IC's) we get

$y^1 = 2.195$, $f^1 = \frac{dy^1}{dx} = 1.8732$ @ $x = 1.1$

$y^2 = 2.3752$, $f^2 = \frac{dy^2}{dx} = 1.7172$ @ $x = 1.2$

Problem 2

2nd order Adams - Moulton method

$$2\frac{dy}{dx} + 3y^2 = 10 \quad , \quad y = 10 \text{ at } x = 1$$

$$\frac{dy}{dx} = 5 - \frac{3y^2}{2} = f(y,x) \quad ; \quad y_0(x_0 = 1) = 10$$

$$f(y_0, x_0) = 5 - 3\frac{(10^2)}{2} = -145$$

Implicit Adams - Moulton (Trapezoidal Method) :

$$y_{n+1} = y_n + \frac{\Delta x}{2}\left[f(y_n, x_n) + f(y_{n+1}, x_{n+1})\right]$$

@ n=0, $y_1 = y_0 + \frac{\Delta x}{2}\left[f(y_0, x_0) + f(y_1, x_1)\right]$

$$y_1 = 10 + \frac{(0.1)}{2}\left[-145 + 5 - 3\frac{y_1^2}{2}\right]$$

$$y_1 = 10 - 7 - 0.075\,y_1^2$$

$$0.075\,y_1^2 + y_1 - 3 = 0$$

Solving using Newton Rhapson : Let $h = 0.075\,y_1^2 + y_1 - 3$

$$h' = 0.15\,y_1 + 1$$

$$y_1^{(k+1)} = y_1^{(k)} - \frac{h(y_1^{(k)})}{h'(y_1^{(k)})}$$

@ k=0, $y_1^{(1)} = y_1^{(0)} - \frac{h(y_1^{(0)})}{h'(y_1^{(0)})}$

estimate using Euler → $y_1^{(0)} = y_0 + \Delta x\, f(y_0) = 10 + (0.1)(-145) = -4.5$

$h(y_1^{(0)}) = h(-4.5) = 0.075(-4.5)^2 + (-4.5) = -5.98125$

$h'(-4.5) = 0.15(-4.5) + 1 = 0.325$

$\therefore y_1^{(1)} = 13.9$

@ k=1,

$h(13.9) = 0.075(13.9)^2 + 13.9 - 3 = 25.39075$

$h'(13.9) = 0.15(13.9) + 1 = 3.085$

$y_1^{(2)} = 13 - \frac{25.39075}{3.085} = 5.67 \rightarrow y_1 \sim 5.67$

Using quadratic formula

$y_1 = \{-15.86, 2.5\}$ ✓ → still far off w/ two iterations but will just use this value

@ n=1 ; $y_2 = y_1 + \frac{\Delta x}{2}\left[f(y_1, x_1) + f(y_2, x_2)\right]$

$f(y_1, x_1) = 5 - \frac{3}{2}y_1^2 = 5 - \frac{3}{2}(5.67)^2 = -43.22$

$y_2 = 5.67 + \frac{\Delta x}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y_2 = 5.67 + \frac{(0.1)}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y = 3.759 + 0.075\,y_2^2$

$0.075\,y^2 + y - 3.759 = 0$

Solving using Newton Rhapson : Let $g = 0.075\,y_2^2 + y_2 - 3.759$

$$g' = 0.15\,y_2 + 1$$

$$y_2^{(k+1)} = y_2^{(k)} - \frac{g(y_2^{(k)})}{g'(y_2^{(k)})}$$

@ k=0, $y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$

d ~ d ✓

@ k=0,

$$y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$$

estimate using Euler → $y_2^{(0)} = y_2 + \Delta x f(y_1) = 5.47 + (0.1)(-43.22) = 1.348$

$g(y_2^{(0)}) = g(1.348) = 0.075(1.348)^2 + (-2.411) = -2.2747$

$g'(1.348) = 0.15(1.348) + 1 = 1.2022$

$\therefore\ y_2^{(1)} = 1.348 - \frac{(-2.2747)}{(1.2022)} = 3.24$

@ k=1,

$g(13.9) = 0.075(3.24)^2 + 3.24 - 3.759 = 0.26832$

$g'(13.9) = 0.15(3.24) + 1 = 1.486$

$y_2^{(2)} = 3.24 - \frac{0.26832}{1.486} = 3.0594 \rightarrow \quad y_2 \approx 3.06$

Using quadratic formula

$y_1 = \{-16.39, 3.0587\}$ ✓ → much better approximation

$\therefore$ For

$x_1 = 1.1,$	$y_1 = 5.47$	w/ $\Delta x = 0.1$
$x_2 = 1.2,$	$y_2 = 3.06$	& given initial conditions

Problem 3

Steady state conduction in a flat plate

$L \Rightarrow$ plate thickness

$\dot{q} \Rightarrow$ heat generation

$$\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q} = 0$$

$k = a + bT$

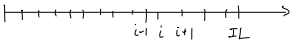
$\dot{q} = d + eT$

$$-k\frac{dT}{dx}\bigg|_{x=0} = h_H\left(T_H - T(x=0)\right)$$

$$-k\frac{dT}{dx}\bigg|_{x=L} = h_L\left(T(x=L) - T_c\right)$$

$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$ ($k_1 = a + bT_1$) $\Big\} F_{BC}^{(1)}(T_1, T_2)$

$-k_2\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$ $\Big\} F_{BC}^{(4)}(T_{IL}, T_{IL-1})$



Discretizing the equation:

$$\left[\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q}\right]_i = \frac{k\frac{dT}{dx}\Big|_{i+\frac{1}{2}} - k\frac{dT}{dx}\Big|_{i-\frac{1}{2}}}{\Delta x} + q_i = 0$$

$$F(T_{i-1}, T_i, T_{i+1}) = \frac{\left(k_{i+\frac{1}{2}}\frac{T_{i+1} - T_i}{\Delta x} - \left(k_{i-\frac{1}{2}}\right)\frac{T_i - T_{i-1}}{\Delta x}\right)}{\Delta x} + d + eT_i = 0 \Big\} F(T_{i+1}, T_i, T_{i-1}) = 0,\ i = 2, 3, \ldots, IL-1$$

$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2} = \frac{(a+bT_i) + (a+bT_{i+1})}{2}$

$k_{i-\frac{1}{2}} = \frac{k_i + k_{i-1}}{2} = \frac{(a+bT_i) + (a+bT_{i-1})}{2}$

$$= \left(\frac{(a+bT_i)}{2(\Delta x)^2} + \frac{(a+bT_{i+1})}{2(\Delta x)^2}\right)(T_{i+1} - T_i) - \left(\frac{a+bT_i}{2(\Delta x)^2} + \frac{a+bT_{i-1}}{2(\Delta x)^2}\right)(T_i - T_{i-1}) + d + eT_i = 0$$

$$= \frac{(a+bT_i)T_{i+1}}{2(\Delta x)^2} + \frac{(a+bT_{i+1})T_{i+1}}{2(\Delta x)^2} - \frac{(a+bT_i)T_i}{2(\Delta x)^2} - \frac{(a+bT_{i+1})T_i}{2(\Delta x)^2} - \frac{(a+bT_i)T_i}{2(\Delta x)^2} - \frac{(a+bT_{i-1})T_i}{2(\Delta x)^2} + \frac{(a+bT_i)T_{i-1}}{2(\Delta x)^2} + \frac{(a+bT_{i-1})T_{i-1}}{2(\Delta x)^2} + d + eT_i = 0$$

$$\frac{\partial F}{\partial T_{i-1}} = -\frac{bT_i}{2(\Delta x)^2} + \frac{bT_{i-1}}{(\Delta x)^2} + \frac{a}{2(\Delta x)^2}$$

$$\frac{\partial F}{\partial T_i} = \frac{bT_{i+1}}{2(\Delta x)^2} - \frac{2bT_i}{(\Delta x)^2} - \frac{a}{(\Delta x)^2} - \frac{(a+bT_{i+1})}{2(\Delta x)^2} - \frac{(a+bT_{i-1})}{2(\Delta x)^2} + \frac{bT_{i-1}}{2(\Delta x)^2} + e = \left(e - \frac{a}{(\Delta x)^2}\right) + \left(-\frac{a}{2(\Delta x)^2} - \frac{T_{i+1}}{2(\Delta x)^2} + \frac{bT_{i+1}}{2(\Delta x)^2}\right) + \left(-\frac{2bT_i}{(\Delta x)^2}\right) + \left(\frac{bT_{i-1}}{2(\Delta x)^2} - \frac{a+bT_{i-1}}{2(\Delta x)^2}\right)$$

$$\frac{\partial F}{\partial T_{i+1}} = \frac{(a+bT_i)}{2(\Delta x)^2} + \frac{a}{2(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2} - \frac{bT_i}{2(\Delta x)^2} = \frac{a}{2(\Delta x)^2} + \frac{bT_i}{2(\Delta x)^2} + \frac{a}{2(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2} - \frac{bT_i}{2(\Delta x)^2} = \frac{a}{(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2}$$

Boundary conditions ($(i) \rightarrow$ initial ; $(f) \rightarrow$ final)

$$-k\left.\frac{dT}{dx}\right|_{x=0} = h_H\left(T_H - T(x=0)\right)$$

$$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$$

$(a+bT_1)$

Initial

$$F_{BC}^{(i)} = \left(\frac{(a+bT_1)T_2 - (a+bT_1)T_1}{\Delta x}\right) + h_H T_H - h_H T_1 = 0$$

$$\frac{\partial F_{BC}^{(i)}}{\partial T_1} = \frac{bT_2}{\Delta x} - \frac{2bT_1}{\Delta x} - h_H - \frac{a}{\Delta x}$$

$$\frac{\partial F_{BC}^{(i)}}{\partial T_2} = \frac{(a+bT_1)}{\Delta x}$$

$$F(T_1, T_2) = (a+bT$$

Newton raphson

$$\begin{bmatrix} \frac{\partial F_1}{\partial T_1} & \frac{\partial F_2}{\partial T_2} \\ \cdots & \cdots \end{bmatrix}$$

$$-k\left.\frac{dT}{dx}\right|_{x=L} = h_L\left(T(x=L) - T_c\right)$$

$$-k_2\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$$

$(a+bT_{IL})$

$$F_{BC}^{(f)} = \left(\frac{(a+bT_{IL})T_{IL} - (a+bT_{IL})T_{IL-1}}{\Delta x}\right) + h_L(T_{IL} - T_c)$$

$$\frac{\partial F_{BC}^{(f)}}{\partial T_{IL}} = \frac{2bT_{IL}}{\Delta x} - \frac{bT_{IL-1}}{\Delta x} + h_L + \frac{a}{\Delta x}$$

$$\frac{\partial F_{BC}^{(f)}}{\partial T_{IL-1}} = -\frac{(a+bT_{IL})}{\Delta x}$$

Therefore to solve second order non-linear equation, we can construct the following $Ax=b$ matrix and use Thomas algorithm to find ΔT values.

$$\begin{bmatrix} \frac{\partial F_{BC}^{(i)}}{\partial T_1} & \frac{\partial F_{BC}^{(i)}}{\partial T_2} & & & \\ & \ddots & & & \\ & \frac{\partial F}{\partial T_{i-1}} & \frac{\partial F}{\partial T_i} & \frac{\partial F}{\partial T_{i+1}} & \\ & & \ddots & & \\ & & \frac{\partial F_{BC}^{(f)}}{\partial T_{IL-1}} & \frac{\partial F_{BC}^{(f)}}{\partial T_{IL}} & \end{bmatrix}_{\vec{T}_{old}} \begin{bmatrix} \\ \Delta\vec{T} \\ \\ \end{bmatrix} = - \begin{bmatrix} \\ \vec{F}(T_{old}) \\ \\ \end{bmatrix}$$

$$\vec{T}_{new} = \vec{T}_{old} + \Delta\vec{T}$$

$$\vec{T}_{old} = \vec{T}_{new} + \Delta\vec{T}$$

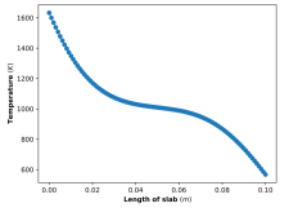
A constraint of $|\Delta\vec{T}| \geq \varepsilon \cdot |\vec{T}_{new}|$ is used, where $\varepsilon = 1e^{-5}$

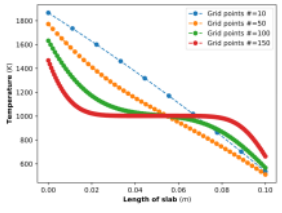
Initial cond. of 1000 K are chosen

The above algorithm is coded in python (w/ the help of Chatgpt to debug the code) and the following plots are obtained

The decrease in temperature (from hot end of ~2000 K) to cold end of ~400 K makes sense since the rod is cooling down. This is also shown by the graphs.

-additionally, the grid spaces were varied from 10 to 150 to give the same results (grid independent)





HW2

Monday, 17 February 2025 6:32 PM

Question 1

a) $\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + au^2$

→ Order = 2
→ Almost linear (au^2 adds non-linearity but all higher-order terms are linear)

b) $u_t + c\,u_x + a\,u^3 = \nu\, u_x\, u_{xx}$

→ Order = 2
→ Quasi-linear (highest order term is multiplied by a lower-order derivative)

c) $u_t + u_x^2 = \nu\, u_{xx}$

→ order = 2
→ Quasi-linear (u_{xx} does not have non-linear coefficients)

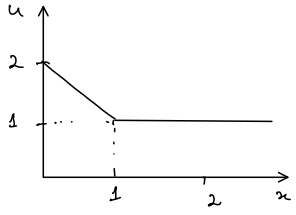
d) $u_t + u_x = \nu\,(u_{xx})^{0.5}$

→ order = 2
→ Non-linear

Question 2 & 4

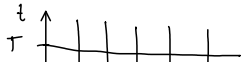
$u_t + u\,u_x = 0$ $\qquad u = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$

$u(x=0) = 2$

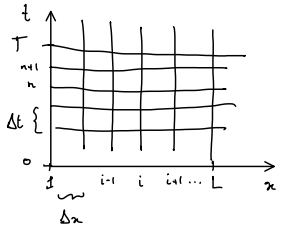


$\frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$; $\frac{2\Delta t}{\Delta x} \le 1$

→ Discretize the domain :



→ Discretize the domain :



→ Discretize the ODE :

time levels : $t^n = n\Delta t$; $n = 0, 1, 2 \ldots$

grid points : $x_i = (i-1)\Delta x$; $i = 1, 2, 3, \ldots m, m+1$

$$u(x_i, t^n)$$

Backward finite difference for u_x : $u_x = \dfrac{u_i^n - u_{i-1}^n}{\Delta x}$

Explicit Euler scheme for u_t : $u_t = \dfrac{u_i^{n+1} - u_i^n}{\Delta t}$

Rearranging the given equation yields :

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}\left[u_i^n \left(u_i^n - u_{i-1}^n\right)\right]$$

Boundary conditions : $u(x=0, t) = 2$

Initial conditions : $u(x, t=0) = F(x) = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$

$\Delta x = \dfrac{L}{nx}$; $nx = 100$; $L = 2$

$\Delta t = 0.001$; $nt = 100$ (number of time steps)

(a) Plot and evaluate solution

Algorithm

↳ First grid parameters are defined based on the stability constraints $\left(\dfrac{2\Delta t}{dx} \le 1\right)$

↳ Spatial grid of x w/ set dx points is initialized ;

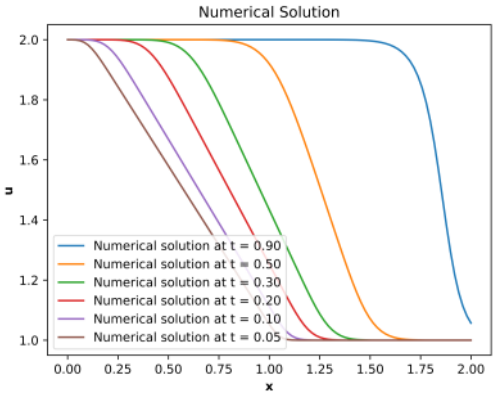
↳ Initial condition for u is established w/ the given function

↳ Nested for loop where outside loop steps the time while inside loop calculates the (n+1) value :

$$u_i^{n+1} = u_i^n - u_i^n \cdot \frac{\Delta t}{\Delta x}\left(u_i^n - u_{i-1}^n\right)$$

↳ The boundary condition is applied, where $u(x=0, t) = 2$

↳ results are plotted for varying time steps ; where $0 < t < 1$



Nonconservative method (w/ the upwind scheme) will not converge to a discont. weak solution as the grid is refined

(b) Plot exact solution via characteristics curves in the x-t plane

Characteristic curve solution → $\frac{dx}{dt} = u(x,t)$; $\underbrace{\frac{du}{dt} = 0}$

u is a constant along the characteristic equation

$x = ut + C_1$

$C_1 = x - ut$

$u = C_2$

Characteristics are straight lines ; C_1 must be a function of u ;

Hence $u = C_2(x - ut)$

$u = F(x - ut)$, where $F(\xi)$ is the provided initial condition @ $t=0$

$x = F(\xi)t + \xi$, where $\xi = \xi(x,t)$ & is the x-intercept of characteristic curve

$0 \leq x \leq 1$; $F(\xi) = 2 - \xi$

$t = \frac{x - \xi}{F(\xi)}$

$x > 1$; $F(\xi) = 1$

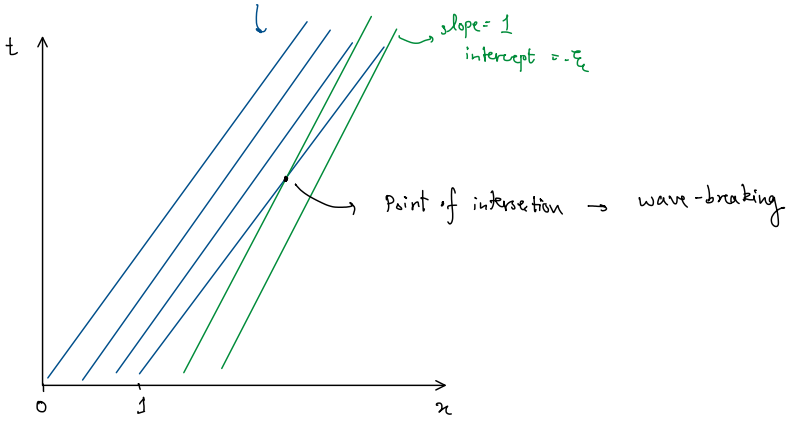
$t = x - \xi$

$t = \frac{x - \xi}{2 - \xi}$

$0 \le \xi < 2 \rightarrow$ positive slope w/ negative t-intercept

$\xi < 0 \rightarrow$ positive slope (less steep) w/ positive intercept

slope $\left(\frac{1}{2-\xi}\right) \rightarrow$



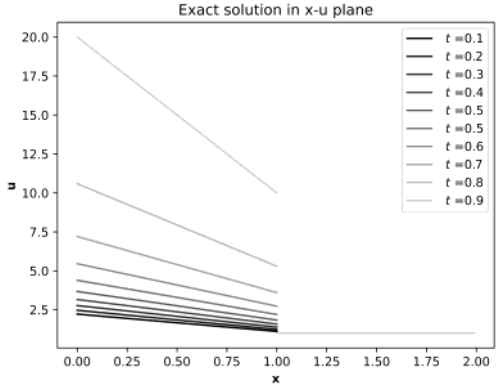
& $u(x,t) = F(x-ut)$

given, $u(x,0) = F(x) = \alpha x + \beta$

$u(x,t) = \alpha(x-ut) + \beta$

$$\boxed{u(x,t) = \frac{\alpha x + \beta}{\alpha t + 1}}$$

For each fixed time, the solution is a straight line w/ slope $\left(\frac{\alpha}{\alpha t+1}\right)$ in the x-u plane



The above graph shows the solution for u where for region

$0 \le x \le 1$; $u(x,t) = F(x-ut) = \frac{2-x}{1-t}$

As t approaches 1, the solution blows up and for varying t the the slope of line is changing i.e. as t increases the slope (and intercept) increases (since $(1-t)$ gets smaller so $\frac{1}{(1-t)}$ gets bigger)

the slope (and intercept) increases (since $(1-t)$ gets smaller so $\frac{1}{(1-t)}$ gets bigger)

$x > 1 \quad ; \quad u(x,t) = 1$

This is just a constant u line for all values of x & t

Therefore $x=1$ forms the discontinuity in the solution

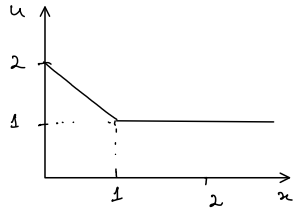
(c) Explain where BC are needed by exact & by numerical solution

For numerical solution, the initial nodes are $u_i^0 = u_0(x_i)$ which is the given expression in the problem. The boundary node is the given $u_0^n = u_1(t^n)$

Question 3 & 4

$$u_t + \frac{\partial(u^2/2)}{\partial x} = 0 \qquad u = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

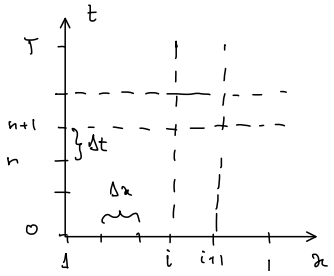
$u(x=0) = 2$

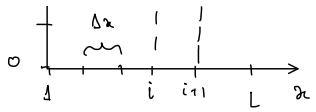


$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} = 0 \qquad ; \quad \frac{2\Delta t}{\Delta x} \le 1$$

(a) Plot and evaluate solution

• Discretize the domain :





- Discretize the ODE:

time levels : $t^n = n\Delta t$; $n = 0, 1, 2 \cdots$

grid points : $x_i = (i-1)\Delta x$; $i = 1, 2, 3 \cdots m, m+1$

$$u_i^{n+1} = u_i^n - \left[\frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} \right]$$

Conservative PDE of the form $u_t + [f(u)]_x = 0$

can be discretized as such:

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x} \left(f(u_i^n) - f(u_{i-1}^n) \right)$$

$nx = 100$ $\quad L = 2$

$dt = 0.001$

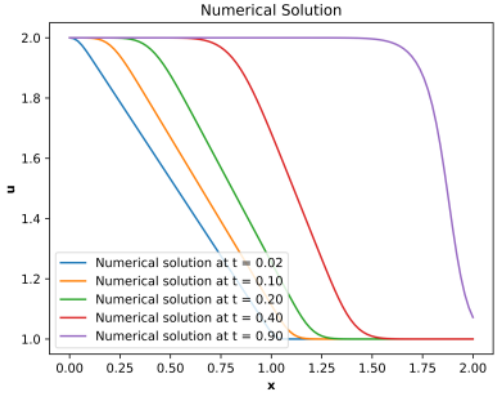
Algorithm:

- A similar algorithm to Q3 is followed w/ some key changes to include the upwind conservative algorithm mentioned above

$$f_plus = \frac{1}{2}(u_i^n)^2$$

$$f_minus = \frac{1}{2}(u_{i-1}^n)^2$$

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}(f_plus - f_minus)$$



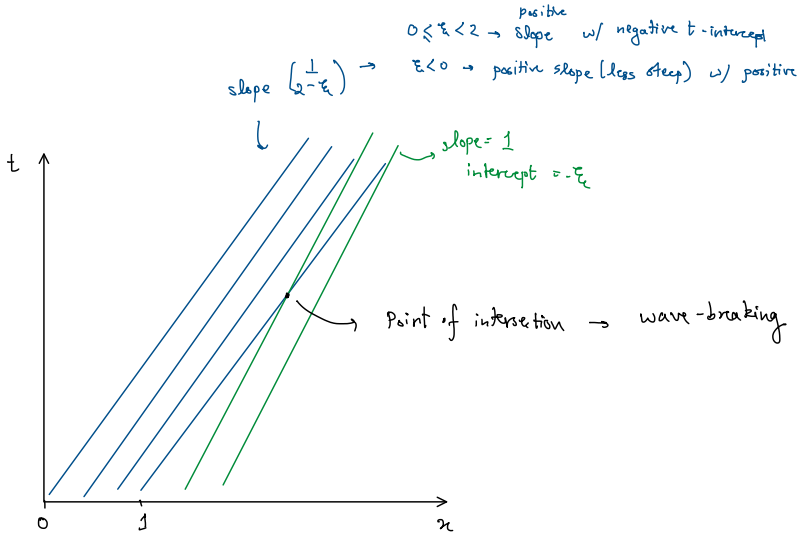
(b) Plot exact solution via characteristics curves in the x-t plane

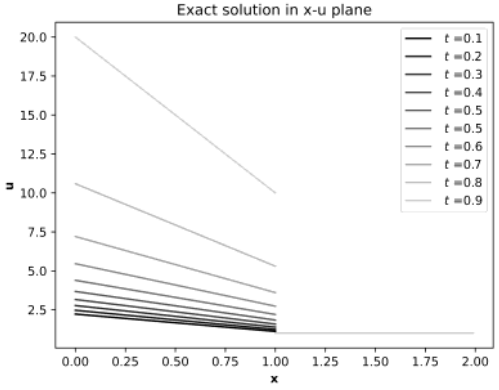
parameterizing by s:

$x(s,0) = s$; $\quad u(s,0) = u_0(s)$

$\frac{\partial x}{\partial t} = u$, $\frac{\partial u}{\partial t} = 0$

$x = u_0(s)\,t + s$ → will be similar to the previous analysis for non-conservative equation. Exact solution is given by the characteristic curves ($x-t$ & $x-u$)





(c) Explain where BC are needed by exact & by numerical solution

The numerical solution uses the same boundary & initial condition nodes: $u_0^n = u_1(t^n)$ & $u_i^0 = u_0(x_i)$

The exact solution gives infinite no. of curves, using the given initial conditions, where a specific solution can be found by using the BC.

Question 4 (contd.)

(a), (b), (c) → see previous parts

(d) Explain weak solution concept. Contrast shock speed

Shocks are a result of intersection of characteristic curves of a given PDE. If the curves are paralell then solution exists for all t. A breaking time of less than 0 can result when the characteristic lines fan out; this results in a global solution for t>0 and a phenomena called rarefaction wave. For initial condition where the condition defined in left domain is larger than than that in the right hand side domain then this would lead to a shock wave where shock speed is the speed of propagation of this wave.

Smooth solutions of conservation laws can break down in a finite amount of time due to intersections in x-t plane (as shown previously). Once a shock occurs, classical solutions are no longer applicable, necessitating a new approach which is termed as weak solution. The involves defining a solution after the breaking point by allowing for u to be discontinuous.

For a weak solution, it must still satisfy some criteria's (i.e. Rankine-Hugonoit jump condition) across the discontinuity curve. This condition gives the shock speed $s = \frac{f(u_L)-f(u_R)}{(u_L-u_R)}$ which is equivalent of saying the ratio of "jump in f(u)" over "jump in u" (from left to right).

(e) Explain entropy condition

There may be cases when the weak solution does not give unique solution after the shock. This is where an entropy condition can be used to disregard non-physical solutions.

The condition allows for a discontinuity only when the left wave is moving faster than the right wave. Therefore, if a wave is propagating with speed s where $s = \frac{dx}{dt}$ then $u^+ < u^-$. This is mathematical equivalent of saying that every physical process the entropy of the system should not decreasing (which is directly related to the 2nd law of thermodynamics).

Question 5 Eigenvalue approach

(a) $5u_{tt} - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

$$A = \begin{bmatrix} 5 & & & \\ & -7 & & \\ & & 8 & \\ & & & 1 \end{bmatrix}$$

Eigenvalue of diagonal matrix are just the diagonals

∴ Hyperbolic
($\lambda_i \neq 0$ & all but one have same sign)

(b) $5u_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

$$A = \begin{bmatrix} 5 & & & \\ & 7 & & \\ & & 8 & \\ & & & 1 \end{bmatrix} \qquad \lambda_i = \{5, 7, 8, 1\}$$

∴ Elliptic
(λ_i all have the same sign)

Question 6 Canonical-form approach ; independent vars $\{x, y, z, t\}$

(a) $5u_{tt} - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Highest order derivative exists for all w/ one different sign → Hyperbolic

(b) $5u_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Highest order derivatives of all vars but all are positive → elliptic

(c) $5u_t - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

highest order derivatives for $\{x, y, z\}$ but t

parabolic in "broad sense"

(d) $xu_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Mixed, need to evaluate determinant of coefficient matrix where the sign of x will determine the property

Question 7

↳ BC's & IC's by 2nd order elliptical, parabolic & hyperbolic PDE's

↳ weak solutions;

Hyperbolic 2nd order PDEs:
Characteristic curves are real and distinct where theres exists domains of influence and dependence for a solution at each node in the domain. Therefore, initial and boundary conditions can be imposed only on where the characteristic base curves enter the domain. The number of conditions is equal to the number of characteristics that can enter the domain at that point. All characteristics passing through the surface need to have their IC/BC defined in order to completely specify conditions for a solution

They are capable of having weak solutions and the nature of these is similar to that of the first order PDE's

Parabolic 2nd order PDEs:
An initial value for the solution must be given at every point of the spatial domain at t=0 and the boundary value for the solution must be given at every point on the boundary of the spatial domain for every instant of time $0 \leq t \leq t_0$ to determine solution up to time t_0

Parabolic do not have weak solutions and any discontinuities are eventually removed so that the solution and its derivatives are all continuous in the domain interior.

Elliptical 2nd order PDEs:
Boundary condition is needed at every point on the domain as these PDEs do not have real characteristics. However, initial conditions are irrelevant since there are no time-like coordinates. These PDEs do not have weak solutions and discontinuities in BC are not propagated into the domain.

Above answers have been paraphrased from Shih-Caughey, Chapter 4

Linear Burger equation : $\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = (1-\theta)\left(\frac{\partial u}{\partial t}\right)_i^n + \theta\left(\frac{\partial u}{\partial t}\right)_i^{n+1}$$

For consistency : $\lim_{\Delta x, \Delta t \to 0} (FDE - PDE)_i^n$

How to check ?

→ Expand terms via Taylor & then substitute

$$u_i^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta t^3}{3!} + \cdots$$

$$u_{i-1}^n = u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{3!} + \cdots$$

$$u_{i+1}^n = u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 + 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t \Delta x^2 + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$u_{i-1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 - \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 - 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t \Delta x^2 - \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = (1-\theta)\left(-c\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} + \nu\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}\right) + \theta\left(-c\frac{u_{i+1}^{n+1} - u_{i-1}^{n+1}}{2\Delta x} + \nu\frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2}\right)$$

$\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t$; $\left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} \cdots$; $\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^4}{4!} \cdots$

doing this for all terms

& getting $\Delta t \to 0$, $\Delta x \to 0$

Question 2

Fourier method of stability analysis (disturbance at every grid) $\varepsilon_i^n = \sum_{-N}^{N} A_j^n e^{I k_j x_i}$

↓

$$\frac{(u_i + \varepsilon)^{n+1} - (u_i + \varepsilon)^n}{\Delta t} = (1-\theta)\left[\cdots\right.$$

Then

$$\varepsilon_i^{n+1} = \sum A_j^{n+1} \exp\left(I k_j x_i\right)$$

$$\varepsilon_i^n = \sum A_j^n \exp\left(I k_j x_i\right)$$

$$\varepsilon_{i+1}^n = \sum A_j^n e^{I k_j x_{i+1}} \qquad x_{i+1} = x_i + \Delta x \qquad x_{i-1} = x_i - \Delta x$$

$\sin\left(I k_j x_{i+1}\right) + \cos\left(I k_j x_{i+1}\right)$

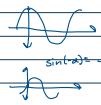
$\cos(-\alpha) = \cos\alpha$

$\exp\left(I k_j \Delta x\right) - \exp\left(-I k_j \Delta x\right) = \sin\left(I k_j x\right)$

$\sin(-\alpha) = -\sin\alpha$

$i\sin(\alpha) + \cos\alpha - \left(i\sin(-\alpha) + \cos(-\alpha)\right)$

$i\sin(\alpha) + \cos\alpha + \sin\alpha - \cos(\alpha)$



$\exp(-I\omega_j \Delta x) - \exp(-\omega_j \Delta x) = \sin(\omega_j x)$

$i\sin(\alpha) + \cos\alpha - (i\sin(-\alpha) + \cos(-\alpha))$

$i\sin(\alpha) + \cos\alpha + \sin\alpha - \cos(\alpha)$

↳ Use this analysis:

expression for A_j^n

↓

define G

↓

find $|G|^2 \rightarrow |x|^2 = x$ (modulus)

$z = x + iy$

$|z|^2 = x^2 + y^2$

$|G| \leq 1$

$|G|^2 \leq 1$

Lax Convergence Theorem

↳ If FDE/FVE's are well-posed, linear PDE's are convergent if the FDE's/FVE's are consistent & stable

↓

→ solution exists & is unique

→ solution depends cont. on the data

Question 1

If $\theta=1$, will FD/FV yield exact solution as $\Delta x, \Delta t \to 0$

Modified equation becomes, $(\theta=1)$:

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = -c\left(\frac{u_{i+1}^{n+1}-u_{i-1}^{n+1}}{2\Delta x}\right) + \nu\,\frac{\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)}{\Delta x^2}$$

To yield exact soln as $\Delta x, \Delta t \to 0$; have to check for convergence (global error goes to 0 as $\Delta x, \Delta t \to 0$)

$$\lim_{\Delta x,\,\Delta t\to 0} N_i^n = U_i^n \quad \text{, where}$$

N_i^n = numerical solution @ (i,n) w/ round off errors

U_i^n = exact solution @ (i,n)

According to Equivalence theorem of Lax,

FDE will be convergent for well-posed, linear PDE's if the FDE is consistent & stable

Showing consistency:

→ Expand terms via Taylor & then substitute

$$u_i^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta t^3}{3!} + \cdots$$

$$u_{i-1}^n = u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^n = u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 + 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t\,\Delta x^2 + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$u_{i-1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 - \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 - 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t\,\Delta x^2 - \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = -c\left(\frac{u_{i+1}^{n+1}-u_{i-1}^{n+1}}{2\Delta x}\right) + \nu\,\frac{\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)}{\Delta x^2}$$

$$\left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = \frac{-c}{2\Delta x}\left[2\left(\left(\frac{\partial u}{\partial x}\right)_i^n \Delta x\right) + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t + \left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \frac{\nu}{\Delta x^2}\left[\left(2u_i + 2\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{1}{3!}\left[2\frac{\partial^3 u}{\partial t^3}\Delta t^3 + 6\frac{\partial^3 u}{\partial t \partial x^2}\Delta t \Delta x^2\right]\right) - 2u_i - 2\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - 2\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} - 2\frac{\partial^3 u}{\partial t^3}\frac{\Delta t^3}{3!}\right]$$

$$\therefore \left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = -c\left[\left(\frac{\partial u}{\partial x}\right)_i^n + \frac{\partial^2 u}{\partial x \partial t}\frac{\Delta t}{2} + \frac{\partial^3 u}{\partial t^2 \partial x}\frac{\Delta t^2}{2} + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{2}\right] + \nu\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + 2\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)\Delta t\right]$$

As $\Delta t, \Delta x \to 0$

$$\therefore \left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = -c\left[\left(\frac{\partial u}{\partial x}\right)_i^n + \frac{\partial^2 u}{\partial x \partial t}\frac{\Delta t}{2} + \frac{\partial^3 u}{\partial t^2 \partial x}\frac{\Delta t^2}{2} + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{2}\right] + \nu\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + 2\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)\Delta t \cdots\right]$$

(each of the Δt / Δx terms → 0)

$$\left(\frac{\partial u}{\partial t}\right)_i^n = -c\left(\frac{\partial u}{\partial x}\right)_i^n + \nu\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n$$

This FDE is equal to the PDE & therefore unconditionally consistent (no Δt or Δx terms remaining & hence independent of the grid & time spacing)

Determining stability:

Fourier transform analysis

$$\frac{(u_i+\epsilon_i)^{n+1}-(u_i+\epsilon_i)^n}{\Delta t} = -c\,\frac{\left((u_{i+1}+\epsilon_{i+1})^{n+1}-(u_{i-1}+\epsilon_{i-1})^{n+1}\right)}{2\Delta x} + \nu\,\frac{\left((u_{i+1}+\epsilon_{i+1})^{n+1}-2(u_i+\epsilon_i)^{n+1}+(u_{i-1}+\epsilon_{i-1})^{n+1}\right)}{\Delta x^2}$$

Subtracting the FDE from the one above, we get:

$$\frac{\epsilon_i^{n+1}-\epsilon_i^n}{\Delta t} = -c\,\frac{\left(\epsilon_{i+1}^{n+1}-\epsilon_{i-1}^{n+1}\right)}{2\Delta x} + \frac{\nu}{\Delta x^2}\left(\epsilon_{i+1}^{n+1}-2\epsilon_i^{n+1}+\epsilon_{i-1}^{n+1}\right)$$

$$\epsilon_i^n = \sum_{j=-N}^{N} A_j^n\, e^{(Ikx_i)}$$

$e^{ix} = \cos x + i\sin x$

$e^{ix} + e^{-ix} = \cos x + i\sin x + \cos x - i\sin x = 2\cos x$

$e^{ix} - e^{-ix} = \cos x + i\sin x - \cos x + i\sin x = 2i\sin x$

Removing the summation & looking at terms separately:

$$\frac{1}{\Delta t}\left(A_j^{n+1}\exp(Ikx_i) - A_j^n\exp(Ikx_i)\right) = \frac{-c}{2\Delta x}\left(A_j^{n+1}\exp\left(Ik(x_i+\Delta x)\right) - A_j^{n+1}\exp\left(Ik(x_i-\Delta x)\right)\right) + \frac{\nu}{\Delta x^2}\left(A_j^{n+1}\exp\left(Ik(x_i+\Delta x)\right) - 2A_j^{n+1}\exp(Ikx_i) + A_j^{n+1}\exp\left(Ik(x_i-\Delta x)\right)\right)$$

$$\Rightarrow \frac{\exp(Ikx_i)}{\Delta t}\left(A_j^{n+1}-A_j^n\right) = \frac{-c}{2\Delta x}A_j^{n+1}\exp(Ikx_i)\left(\exp(Ik\Delta x) - \exp(-Ik\Delta x)\right) + \frac{\nu}{\Delta x^2}A_j^{n+1}\exp(Ikx_i)\underbrace{\left[\exp(Ik\Delta x) - 2 + \exp(-Ik\Delta x)\right]}_{2\cos(k\Delta x)-2}$$

$$\Rightarrow \left(A_j^{n+1}-A_j^n\right) = \frac{-c\Delta t}{2\Delta x}A_j^{n+1}2I\sin(k\Delta x) + \frac{\nu\Delta t}{\Delta x^2}A_j^{n+1}2\left[\cos(k\Delta x)-1\right]$$

$$C = \frac{c\Delta t}{\Delta x} \quad ; \quad d = \frac{\nu \Delta t}{\Delta x^2} \quad ; \quad G = A_j^{n+1} / A_j^n$$

$$G - 1 = G\left(-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right)$$

$$G = \frac{1}{\left[1 + IC\underbrace{\sin(k_j\Delta x)}_{\alpha} + 2d\underbrace{(1-\cos k_j\Delta x)}_{\beta}\right]}$$

$$G = \frac{1}{1 + I\alpha + \beta}$$

$$\frac{1}{[(1+\beta)+I\alpha]} \cdot \frac{(1+\beta)-I\alpha}{(1+\beta)-I\alpha} = \frac{(1+\beta)-I\alpha}{(1+\beta)^2 - \alpha^2}$$

$|G| \le 1$ or $|G|^2 \le 1$

$$|G|^2 = \frac{(1+\beta)^2}{\left((1+\beta)^2-\alpha^2\right)^2} + \frac{\alpha^2}{\left((1+\beta)^2-\alpha^2\right)^2}$$

$$|G|^2 = \frac{(1+\beta)^2 + \alpha^2}{\left((1+\beta)^2-\alpha^2\right)^2} \le 1$$

$$(1+\beta)^2 + \alpha^2 \le \left((1+\beta)^2 - \alpha^2\right)^2$$

$$(1+\beta)^2 + \alpha^2 \le (1+\beta)^4 + \alpha^4 - 2(1+\beta)^2\alpha^2$$

$$-(1+\beta)^4 + (1+\beta)^2(+2\alpha^2+1) - \alpha^4 + \alpha^2 \le 0$$

$$(1+\beta^2)\left[-(1+\beta)^2 + (2\alpha^2+1)\right] - \alpha^2(\alpha^2+1) \le 0$$

$\alpha = C\sin(k_j\Delta x)$

$\beta = (1-\cos(k_j\Delta x)) \ge 0$

Only for certain values of α, C, d will the above inequality be satisfied

↳ Hence conditionally stable

↳ Hence we cannot say if for $\theta = 1$ the given FD will always give an exact solution

Question 2

Dissipation error → analyzed via Fourier Method

$$\left(|G_{PDE}|^n - |G_{FDE}|^n\right) A_j^0$$

↑ disturbance introduced at the zeroth time scale

Finding $|G_{PDE}|$:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \quad \text{— differentiating}$$

$$\Rightarrow a_j e^{a_j t} e^{Ik_j x} + c\, e^{a_j t} e^{Ik_j x}(Ik_j) = \nu\left((Ik_j)^2 e^{a_j t} e^{Ik_j x}\right)$$

$$\therefore \left(a_j + cIk_j + \nu k_j\right) = 0$$

$$a_j = -\nu k_j - (ck_j)I$$

$$\therefore a = -\nu k_j \quad ; \quad b = (-ck_j)$$

$$G_{PDE} = \frac{u_j(x, t+\Delta t)}{u_j(x,t)} = e^{a_j \Delta t} = e^{(a+Ib)\Delta t} = e^{-\nu k_j^2 \Delta t} \cdot e^{-cIk_j \Delta t}$$

$$|G_{PDE}| = e^{-\nu k_j^2 \Delta t}$$

Finding $|G_{FDE}|$: (for $\theta = 1/2$)

Given FDE, we can use the fourier method to arrive at the following expression for $\theta = \frac{1}{2}$

$$\frac{(u_i+\varepsilon)^{n+1} - (u_i+\varepsilon)^n}{\Delta t} = \frac{1}{2}\left[-\frac{c\left((u_{i+1}+\varepsilon)^n - (u_{i-1}+\varepsilon)^n\right)}{2\Delta x} + \frac{\nu\left((u_{i+1}+\varepsilon)^n - 2(u_i+\varepsilon)^n + (u_{i-1}+\varepsilon)^n\right)}{\Delta x^2}\right]$$

$$+ \frac{1}{2}\left[-\frac{c\left((u_{i+1}+\varepsilon)^{n+1} - (u_{i-1}+\varepsilon)^{n+1}\right)}{2\Delta x} + \frac{\nu\left((u_{i+1}+\varepsilon)^{n+1} - 2(u_i^{n+1}+\varepsilon) + (u_{i-1}+\varepsilon)^{n+1}\right)}{\Delta x^2}\right]$$

Subtracting from the original FDE & using fourier components we get:

$$\left[\varepsilon_i^n = \sum_{j=1}^{M} A_j^n \exp\left(Ik_j x_i\right)\right] \quad ;$$

$$\left(A_j^{n+1} - A_j^n\right)\exp(Ik_j x_i) = \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x}A_j^n \exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2}A_j^n \exp\left[Ik_j x_i\right]\left[1-\cos(k_j\Delta x)\right]\right]$$

$$+ \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x}A_j^{n+1}\exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2}A_j^{n+1}\exp\left(Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right]\right]$$

Defining $\frac{c\Delta t}{\Delta x} = C$ & $\frac{\nu\Delta t}{\Delta x^2} = d$ & $G = \frac{A_j^{n+1}}{A_j^n}$,

$$G - 1 = \frac{1}{2}\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right] + \frac{1}{2}G\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right]$$

$$\therefore G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}$$

$$G = \frac{\left[1 - \underbrace{d(1-\cos k_j\Delta x)}_{a}\right] + \underbrace{IC\sin(k_j\Delta x)}_{Im}}{\underbrace{1 + d(1-\cos k_j\Delta x)}_{b} + IC\sin(k_j\Delta x)}$$

$$= \frac{a + Im}{b + Im}$$

$$= \frac{(a+Im)(b-Im)}{b^2+m^2} = \frac{ab - a(Im) + b(Im) + m^2}{b^2+m^2}$$

$$= \frac{ab + Im(b-a) + m^2}{b^2+m^2}$$

$$G = \frac{ab+m^2}{(b^2+m^2)} + I\left(\frac{m(b-a)}{(b^2+m^2)}\right) \qquad ; \quad |z| = \sqrt{x^2+y^2} \quad z = x+iy$$

$$|G|^2 = \left(\frac{ab+m^2}{(b^2+m^2)}\right)^2 + \left(\frac{m(b-a)}{(b^2+m^2)}\right)^2$$

$$|G_{FDE}| = |G| = \left[\frac{(ab+m^2)^2 + m^2(b-a)^2}{(b^2+m^2)^2}\right]^{1/2}$$

$\therefore$ For an initial disturbance of $A_0 = 1$

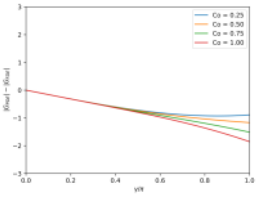
$$(|G_{PDE}|^n - |G_{FDE}|^n) = e^{-\nu k_j^2 \Delta t} - \left[\frac{(ab+m^2)^2 + m^2(b-a)^2}{(b^2+m^2)^2}\right]^{1/2}$$

Plot below for varying C values to see how the above absolute error behaves

1) Error increases quickly as C increases

2) Fourier components w/ the smallest wave no. (longest wavelength) are dissipated the least

3) Stable numerical solutions For smaller Co numbers



Question 3

Dispersion error

Finding ϕ_{PDE} first:

$$u_j(x,t) = e^{\alpha_j t} e^{Ik_j x} \qquad ; \quad \alpha_j = a + Ib$$

$$\left.\begin{aligned} \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} &= \nu \frac{\partial^2 u}{\partial x^2} \end{aligned}\right\} \text{ differentiating}$$

$$\Rightarrow \alpha_j e^{\alpha_j t} e^{Ik_j x} + c e^{\alpha_j t} e^{Ik_j x} (Ik_j) = \nu\left((Ik_j)^2 e^{\alpha_j t} e^{Ik_j x}\right)$$

$$= (\alpha_j + cIk_j + \nu k_j) = 0$$

$$\alpha_j = -\nu k_j - (ck_j)I$$

$$\therefore a = -\nu k_j \qquad ; \quad b = (-ck_j)$$

$$G_{PDE} = \frac{u_j(x,t+\Delta t)}{u_j(x,t)} = e^{\alpha_j \Delta t} = e^{(a+ib)\Delta t} = e^{-\nu k_j^2 \Delta t} \cdot e^{-ck_j I \Delta t}$$

$$\phi_{PDE} = -ck_j \Delta t$$

Given FDE, we can use the fourier method to arrive at the following expression for $\theta = \frac{1}{2}$

$$\frac{(u_i+\varepsilon)^{n+1} - (u_i+\varepsilon)^n}{\Delta t} = \frac{1}{2}\left[\frac{-c(u_{i+1}+\varepsilon)^n - (u_{i-1}+\varepsilon)^n}{2\Delta x} + \frac{\nu(u_{i+1}+\varepsilon)^n - 2(u_i+\varepsilon)^n + (u_{i-1}+\varepsilon)^n}{\Delta x^2}\right]$$

$$+ \frac{1}{2}\left[\frac{-c(u_{i+1}+\varepsilon)^{n+1} - (u_{i-1}+\varepsilon)^{n+1}}{2\Delta x} + \frac{\nu(u_{i+1}+\varepsilon)^{n+1} - 2(u_i^{n+1}+\varepsilon) + (u_{i-1}+\varepsilon)^{n+1}}{\Delta x^2}\right]$$

Subtracting from the original FDE & using fourier components we get:

$$\left[\varepsilon_i^n = \sum_{j=-N}^{N} A_j^n \exp\left[Ik_j x_i\right]\right] \qquad ;$$

$$(A_j^{n+1} - A_j^n)\exp(Ik_j x_i) = \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x} A_j^n \exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2} A_j^n \exp\left[Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right]\right]$$

$$+ \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x} A_j^{n+1}\exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2} A_j^{n+1}\exp\left(Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right)\right]$$

Defining $\frac{c\Delta t}{\Delta x} = C$ & $\frac{\nu\Delta t}{\Delta x^2} = d$ & $G = \frac{A_j^{n+1}}{A_j^n}$,

$$G - 1 = \frac{1}{2}\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right] + \frac{1}{2}G\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right]$$

$$\therefore \quad G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}$$

$$G = \left[\overbrace{1 - \tfrac{1}{2}(\ldots)}^{a}\right] - I\overbrace{C\sin(k_j\Delta x)}^{m}$$

$$\therefore \quad G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)\right] + IC\sin(k_j\Delta x)}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)\right] + IC\sin(k_j\Delta x)}$$

$$G = \frac{\underbrace{\left[1 - d(1-\cos k_j\Delta x)\right]}_{a} + \overbrace{IC\sin(k_j\Delta x)}^{m}}{\underbrace{1 + d(1-\cos k_j\Delta x)}_{b} + IC\sin(k_j\Delta x)}$$

$$= \frac{a + Im}{b + Im}$$

$$= \frac{(a+Im)(b-Im)}{b^2+m^2} = \frac{ab - a(Im) + b(Im) + m^2}{b^2+m^2}$$

$$= \frac{ab + Im(b-a) + m^2}{b^2+m^2}$$

$$G = \frac{ab+m^2}{(b^2+m^2)} + I\left(\frac{m(b-a)}{(b^2+m^2)}\right)$$

$$\phi_{FDE} = \tan^{-1}\left[\frac{Im_G}{Re_G}\right] = \tan^{-1}\left[\frac{m(b-a)}{(b^2+m^2)} \cdot \frac{(b^2+m^2)}{(ab+m^2)}\right]$$

$$= \tan^{-1}\left[\frac{m(b-a)}{(ab+m^2)}\right]$$

$$\therefore \quad \boxed{\phi_{FDE} = \tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]}$$

$$b-a = (1 + d - d\cos\theta) - 1 + d - d\cos\theta$$
$$= 2d - 2d(\cos\theta)$$
$$= 2d(1-\cos\theta)$$

$$ab = (1-d+d\cos\theta)(1+d-d\cos\theta)$$
$$= 1 + d\cos\theta - d - d^2 + d^2\cos\theta + d\cos\theta + d^2\cos\theta - d^2\cos^2\theta$$
$$= 1 - d^2 + 2d^2\cos\theta - d^2\cos^2\theta$$
$$= 1 - d^2(1 - 2\cos\theta) - d^2\cos^2\theta$$

Dispersion errors for each component:

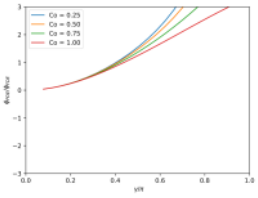
(Absolute error) $\quad \varepsilon(\phi_{PDE} - \phi_{FDE}) = \varepsilon\left(ck_j\Delta t - \tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]\right)$

Relative errors:

$$\Theta = \frac{\phi_{FDE}}{\phi_{PDE}} = \frac{\tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]}{ck_j\Delta t}$$

Dependent on both Δt & Δx

Plot graphs



The relative phase error is decreases with increasing Co number

Question 4

1. **What is the difference between a finite-difference (FD) method and a finite-volume (FV) method?**
   a. Finite difference uses propertues of Taylor expansion to approximate partial derivatives in a given PDE. The domain (where a PDE solution needs to be obtained) is discretized to obtain a mesh/grid where local solution at each grid point is obtained via solving system of linear equations obtained after discretization.
   b. Finite volume converts PDE's to integral forms and solves over a set of small cells that form the control volumes; hence, the domain is discretized to control volumes and the flux of interested quantities are evaluated at the volume boundaries. FVM, thus, uses cell-averaged values. FVM method is also intrinsically conservative, where the conservative laws (in - out + gen = accumulation) are held for each discrete volume.

2. **Contrast approximation of derivatives versus approximating fluxes.**
   Derivatives provide local pointwise information (I.e local changes in a function) and usually approximated using Taylor series; whereas approximating fluxes is when the governing equations are integrated over a finite control volume; flux approximations also ensure conservation.
3. **What is a cell-centered method? What is a vertex-centered method?**
   For a given mesh for FVD analysis, the mesh can be defined in these two given methods:
   a. Cell-centered is a FV discretization method that consists of selecting mesh points as the cell faces that are labeled at half-integer values ($i \pm 1/2$) and the function values are defined at the centers of these cells i.e. the unknowns are the centers of the mesh cells and the grid lines define the finite volume and surfaces
   b. Vertex centered method, in contrast, is when the unknowns are defined at the corners of the mesh. This provides a larger flexibility for definition of control volumes
4. **Define conditional and unconditional consistency. Why is conditional consistency a concern?**
   a. Consistency is defined when the solution for a given FDE matches the exact solution locally, as time and space steps tend to zero. Conditional consistency means that the consistency of the FDE does matter with the rate of dx and dt going to zero, whereas unconditional consistency means that the defined FDE is consistent regardless of how the time and space steps go to zero. Conditional consistency could be a concern because the FDE solution is very sensitive to reductions in dx and dt and it is possible to get inaccurate answers as these are refined. This makes grid sensitivity and independence study quite complex.
5. **Define conditional and unconditional stability. Define unconditionally unstable.**
   a. Stability is concerned with numerical stability of the FDE solution where a solution is stable when the stability error (exact solution of FDE without round off errors - numerical solution with roundoff errors) tends to zero (or bounded) as the number of time steps (n) increases
   b. Conditional stability means that for only for a given inequality involving a critical parameter (function of dt and/or dx) the solution is stable. Unconditional stability means that the solution is stable for all values of dt and dx. Unconditional unstable means that for no values of a given parameter is the solution stable. (I.e. unstable for all of domain since the stability condition is never satisfied)
6. **When should one use explicit methods? When should one use implicit methods?**
   a. Implicit methods are expressions where the derivatives at different mesh points appear simaltaneously to give coupled equations for solving FD method. Their advantage is in general higher order of accuracy as derivates/FDs at different mesh points are related to each other. Additionally, for a linear PDE, implicit methods generally have unconditional stability. The penalty of implicit solutions can be computational cost
   b. Explicit on other hand does not give coupled FD equations (and hence computation can be effectively parallelized).
7. **In general, which Fourier component is the most unstable?**
   a. High wavenumbers modes (high frequency) are the most unstable fourier components due to various reasons: one of them is aliasing error and higher dispersion, dissipation errors
8. **8. What is the physical meaning of the modified equation of a FD/FVE?**
   a. Modified equation is derived from taylor series expression helps identify the truncation errors, dispersion and dissipation effects. For example, the modified equation can tell us of the order of dissipation based on the lowest even order derivative term in the FDE.
9. **The Dufort-Frankel method is an explicit method. But, it is unconditionally stable and conditionally consistent. Comment**

on how to do mesh and time-step refinement.

- **a.** The method introduces time averaging term which does not match the original behavior as dt goes to 0. Therefore dx and dt need to be refined together and dt is generally on the order of dx^2 to help maintain consistency and accuracy.

10. **What is numerical dispersion**
    - **a.** Dispersion occurs when numerical method introduces phase speed errors which results in waves with different wavelentghs being propagated at incorrect speed leading to a distorted wave solution.
    - **b.** Error in the phase of the solution is defined as dispersion error
11. **Why can we use the Fourier method to analyze numerical dissipation and numerical dispersion?**
    Fourier expansions allows a quantitative estimation of errors and the decomposition separates time and space dependence of the given FDE/modified equation. The time behavior is contained in the amplitudes while the fourier modes contain the full space dependence. Therefore both dissipation and dispersion (relating to the phase) can be analyzed.
12. **Define convergence.**
    - **a.** The numerical solution must tend to the exact solution of the mathematical model (PDE) as time and space steps tend to zero
13. **Define the Lax equivalence theorem.**
    - **a.** FDEs/FVEs of well-posed, linear PDEs are convergent if the they are consistent and stable i.e. consistency and stability are necessary and sufficient conditions to show convergence under the assumptions mentioned
14. **Define conservative error.**
    - **a.** Error introduced when FD scheme fails to preserve a conserved quantity (for example entropy)
15. **Define transportive error.**
    - **a.** Occurs when a FD/FV scheme gives incorrect speed of wave leading to incorrect phase speed and wave distortion.
16. **Define aliasing error.**
    - **a.** This error results from inability of a discrete grid to effectively represent a continuous function, where for instance, the amplitude of a resolvable wavelength is changed due to choice of grid spacing. This error mainly ocurs for PDEs that are non-linear, where fourier components multiply each other and create new components that have either longer or shorter wavelengths.
17. **What is the spatial resolution of a mesh?**
    - **a.** Choosing grid spacing that allows the solution to model the physics (*i.e.* choosing appropriate dx and having enough number of grid points) but also optimizing grid points efficiency (to reduce computation time). Hence we use physics driven understanding to choose length scales that must be resolved.
18. **What is the spatial resolution of a FD/FV equation on a mesh?**
    - **a.** Discretization and replacing PDE with FDE results in additional reduction in spatial resolution
19. **What is the numerical connection between elliptic PDEs and parabolic and hyperbolic PDEs? Hint: contrast iterative space and time**
    - **a.** Elliptical PDE is solved as a global system using iterations to converge to a constant value usually via methods such as Jacobi or Gauss Seidel. These usually do not have explicit time dependence
    - **b.** Parabolic and hyperbolic PDEs are solved using iterative time step and solved using either implicit or explicit techniques

Sorry couldn't screenshot the code for some reason the image quality was getting really bad; will fix for next hw

Question 2 Code:

```
def diss_err(Co):
    # Inputs:

    dx = 0.5

    dt = dx/2
    c = Co*dx/dt
    v = 1
    d = v*dt/dx**2


    k = np.linspace(0,10,100)
    theta = k*dx/np.pi
    #x_axis = theta/np.pi

    a = 1-d*(1-np.cos(theta))
    b = 1+d*(1-np.cos(theta))
    m = Co*np.sin(theta)

    G_PDE = np.exp(-v*k*dt)
    G_FDE = np.sqrt(((a*b + m**2)**2 + m**2*(b-a)**2)/(b**2 + m**2)**2)

    err_diss = (G_PDE - G_FDE)/G_PDE
    plt.plot(theta,err_diss,label="Co = {:.2f}".format(Co))
    #

Co=np.array([0.25,0.5,0.75,1])
for i in Co:
    diss_err(i)

plt.xlabel(r" $ \gamma / \pi $",weight="bold")
plt.ylabel(r"$ |G_{PDE}| - |G_{FDE}| $");
plt.xlim([0,1])
plt.ylim([-3,3])
plt.legend()

plt.savefig("G_graph.png",dpi=800,bbox_inches = "tight")
```

Question 3 code:

```
def disp_err_rel(Co):
    # Dispersion error for each component (relative)

    dx = 0.25

    dt = dx/2
    c = Co*dx/dt
    v = 1
    d = v*dt/dx**2


    k = np.linspace(0.1,5,100)
    theta = k*np.pi*dx

    a = 1-d*(1-np.cos(theta))
    b = 1+d*(1-np.cos(theta))
    m = Co*np.sin(theta)

    phi_PDE = c*k*dt
    phi_FDE = np.arctan((Co*np.sin(theta)*2*d*(1-np.cos(theta)))/(1-d**2*(1-2*np.cos(theta))-d**2
*(np.cos(theta))**2+Co**2*(np.sin(theta))**2))

    err_phase = phi_FDE/phi_PDE
    plt.plot(theta,err_phase,label="Co = {:.2f}".format(Co))

Co=np.array([0.25,0.5,0.75,1])
for i in Co:
    disp_err_rel(i)

plt.xlabel(r" $ \gamma / \pi $",weight="bold")
plt.ylabel(r"$\phi_{FDE}/\phi_{FDE}$");
plt.xlim([0,1])
plt.ylim([-3,3])
plt.legend()

plt.savefig("phi_graph.png",dpi=800,bbox_inches = "tight")
```

HW5

Monday, 24 March 2025 2:34 PM

Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = \frac{\partial^2 u}{\partial y^2}$$

$(x, y, t) \rightarrow (\xi, \eta, \tau)$

(a) We can use coordinate transformations: (using $t = \tau$ & $x \neq f(x,y)$)

$$\frac{\partial}{\partial x} = \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial y} = \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}$$

term 1 term 2 term 3 term 4

$$\therefore \boxed{\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right) u + \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right) u^2 + \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right) uv = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)^2 u}$$

Chain-Rule Conservation-Law form

(b) To maintain conservation have to convert to Strong Conservation-Law form

1) $J \times$ (PDE from above)

2) Move J term inside via chain rule

Term 1:

$$\therefore J\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right) u = J\frac{\partial u}{\partial \tau} + J\xi_t \frac{\partial u}{\partial \xi} + J\eta_t \frac{\partial u}{\partial \eta}$$

$$= \left(\frac{\partial (Ju)}{\partial \tau} - u\frac{\partial J}{\partial \tau}\right) + \frac{\partial (J\xi_t u)}{\partial \xi} - u\frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - u\frac{\partial (J\eta_t)}{\partial \eta}$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(u\frac{\partial J}{\partial \tau} + u\frac{\partial (J\xi_t)}{\partial \xi} + u\frac{\partial (J\eta_t)}{\partial \eta}\right)$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \underbrace{\left(\frac{\partial J}{\partial \tau} + \frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t)}{\partial \eta}\right)}_{0 \text{ (geometric conservation law)}} u$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi}^{0} + \frac{\partial (J\eta_t u)}{\partial \eta}^{0} = \frac{\partial (Ju)}{\partial \tau}$$

Similar manipulation of term 2, & term 3 (first order spatial derivatives)

term 2:

$$J\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right) u^2 = \frac{\partial (J\xi_x u^2)}{\partial \xi} + \frac{\partial (J\eta_x u^2)}{\partial \eta} - u^2 \underbrace{\left(\frac{\partial (J\xi_x)}{\partial \xi} + \frac{\partial (J\eta_x)}{\partial \eta}\right)}_{=0 \text{ (geometric conservation law)}}$$

term 3:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right) uv = \frac{\partial (J\xi_y uv)}{\partial \xi} + \frac{\partial (J\eta_y uv)}{\partial \eta} - uv \underbrace{\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\eta_y)}{\partial \eta}\right)}_{=0 \text{ (geometric cons. law)}}$$

term 4:

$$J\frac{1}{J}\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)^2 u = \frac{1}{J}\left(J\xi_y \frac{\partial}{\partial \xi} + J\eta_y \frac{\partial}{\partial \eta}\right)^2 u$$

$$= \frac{1}{J}\left(J^2\xi_y^2 \frac{\partial^2}{\partial \xi^2} + J^2\eta_y^2 \frac{\partial^2}{\partial \eta^2} + 2J^2\xi_y\eta_y \frac{\partial^2}{\partial \xi \partial \eta}\right) u$$

$$= \frac{1}{J}\left(\frac{\partial}{\partial \xi}(J^2\xi_y^2 u) - u\frac{\partial}{\partial \xi}(J^2\xi_y^2) + \frac{\partial}{\partial \eta}(J^2\eta_y^2 u) - u\frac{\partial}{\partial \eta}(J^2\eta_y^2) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y u) - 2u\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y)\right)$$

$$= \frac{1}{J}\left[\frac{\partial^2}{\partial \xi^2}(J^2\xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2\eta_y^2 u) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y u)\right] - \frac{u}{J}\underbrace{\left(\frac{\partial^2}{\partial \xi^2}(J^2\xi_y^2) + \frac{\partial^2}{\partial \eta^2}(J^2\eta_y^2) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y)\right)}_{0 \text{ (= geometric conservation law)}}$$

Therefore,

$J\left[\frac{\partial \xi}{\partial x} \quad \frac{\partial \xi}{\partial y} \quad \frac{\partial \eta}{\partial x} \quad \frac{\partial \eta}{\partial y}\right]$ $J\left[\frac{\partial}{\partial \xi}\left(\frac{\partial x}{\partial \eta}\right) \quad \frac{\partial}{\partial \eta}\left(\frac{\partial x}{\partial \eta}\right) \quad \frac{\partial}{\partial \xi}\left(\frac{\partial x}{\partial \xi}\right)\right]$

0 (= geometric conservation law)

Therefore,

Final strong conservative form:

$$\frac{\partial (Ju)}{\partial \tau} + \left[\frac{\partial}{\partial \xi}(J\xi_x u) + \frac{\partial}{\partial \eta}(J\eta_x u)\right] + \left[\frac{\partial}{\partial \xi}(J\xi_y u) + \frac{\partial}{\partial \eta}(J\eta_y u)\right] = \frac{\partial^2}{\partial \xi^2}(J^2 \xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2 \eta_y^2 u) + \frac{\partial^2}{\partial \xi \partial \eta}(J^2 \xi_y \eta_y u)$$

$$\boxed{\frac{\partial (Ju)}{\partial \tau} + \frac{\partial}{\partial \xi}(J\xi_x u^2 + J\xi_y uv) + \frac{\partial}{\partial \eta}(J\eta_x u^2 + J\eta_y uv) = \frac{\partial^2}{\partial \xi^2}(J^2 \xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2 \eta_y^2 u) + \frac{\partial^2}{\partial \xi \partial \eta}(J^2 \xi_y \eta_y u)}$$

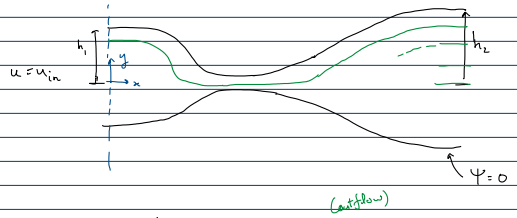
(c) Highlighted terms in blue show the 4 geometric conservation laws (in expression above)

Problem 2

Incompressible potential flow $\nabla^2 \psi = 0$ (ψ: stream function)

(a) Describe the problem

(b) Governing equations



$u = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \psi}{\partial x}$

2D, axisymmetric flow ; ψ = stream function (constant along a stream line)

$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$

governing equations

$\frac{\partial}{\partial x}(-v) + \frac{\partial u}{\partial y} = 0$

$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0$

BC's:

(Inflow) $u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$

(outflow) $\left.\frac{\partial \psi}{\partial x}\right|_{x=c} = 0$

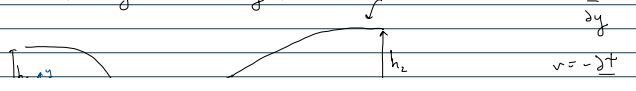
(walls) $\psi = 0$ (@ curve 1)

$\psi = u_{in} h_1$ (@ curve 2)

$$\int_{(y-h_1)}^{(y+h_1)} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

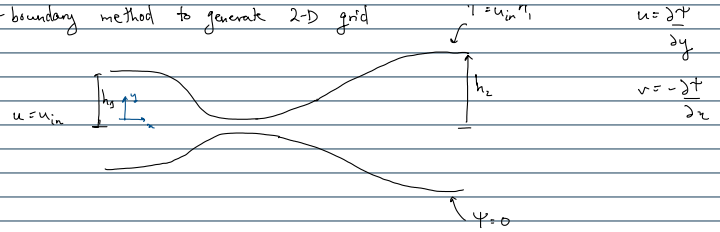
$$\psi(y) = u_{in}\left((y+h_1) - (y-h_1)\right) = u_{in}(2h_1)$$

(c) Two-boundary method to generate 2-D grid



$u = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \psi}{\partial x}$

(c) Two-boundary method to generate 2-D grid



$$u = \frac{\partial \psi}{\partial y} \qquad v = -\frac{\partial \psi}{\partial x}$$

$$u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$$

$$\int_{-h_1}^{h_1} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

(d) Transform (x, y) to (ξ, η)

$$\nabla^2 \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$J^2 \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta} \right)^2 \psi + J^2 \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta} \right)^2 \psi = 0$$

Strong Conservative

$$\Rightarrow \left[\frac{\partial^2 (J^2 \xi_x^2 \psi)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_x^2 \psi)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_x \eta_x^2 \psi\right) \right] + \left[\frac{\partial^2 (J^2 \xi_y^2 \psi)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_y^2 \psi)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_y \eta_y^2 \psi\right) \right]$$

$$- \psi \left[\frac{\partial^2 (J^2 \xi_x^2)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_x^2)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_x \eta_x^2\right) + 2\frac{\partial}{\partial \eta}\left(J^2 \xi_y^2 \eta_y^2\right) \right] = 0 \ ?$$

$$\Rightarrow \frac{\partial^2}{\partial \xi^2}\left(J^2 \xi_x^2 \psi + J^2 \xi_y^2 \psi\right) + \frac{\partial^2}{\partial \eta^2}\left(J^2 \eta_x^2 \psi + J^2 \eta_y^2 \psi\right) + 2\frac{\partial}{\partial \xi \partial \eta}\left(J^2 \xi_x \eta_x^2 + J^2 \xi_y^2 \eta_y^2\right) = 0$$

y_n^2 $\qquad \psi_{i+1} - 2\psi + \psi_{-1}$

$y_n = \xi_x \cdot J$

(e), (f) FDE/FVE's for governing equations + algorithm

$$\frac{\partial^2 \psi}{\partial \xi^2} = \frac{\psi_{i+1,j} - 2\psi_{ij} + \psi_{i-1,j}}{\Delta \xi^2}$$

$$\frac{\partial^2 \psi}{\partial \eta^2} = \frac{\psi_{i,j+1} - 2\psi_{ij} + \psi_{i,j-1}}{\Delta \eta^2}$$

$$\xi_x = \frac{\partial \xi}{\partial x} = \frac{y_\eta}{J}$$

$$\frac{\partial^2 \psi}{\partial \xi \partial \eta} = \frac{\psi_{i+1,j+1} - \psi_{i+1,j-1} - \psi_{i-1,j+1} + \psi_{i-1,j-1}}{4 \Delta \xi \Delta \eta}$$

(g), (h) Code + solutions for velocity = 100 m/s

(i) Discuss accuracy of results.

HW5 final

Monday, 24 March 2025 2:34 PM

Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = \frac{\partial^2 u}{\partial y^2} \qquad (x,y,t) \Leftrightarrow (\xi,\eta,\tau)$$

(a) We can use coordinate transformations: (using $t=\tau$ & $\tau \neq f(x,y)$)

$$\frac{\partial}{\partial x} = \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial y} = \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}$$

term 1 term 2 term 3 term 4

$$\therefore \boxed{\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right)u + \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)u^2 + \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)uv = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right)}$$

Chain-Rule Conservation-Law form

(b) To maintain conservation have to convert to Strong Conservation-Law form

1) $J \times$ (PDE from above)

2) Move J term inside via chain rule

Term 1:

$$\therefore J\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right)u = J\frac{\partial u}{\partial \tau} + J\xi_t \frac{\partial u}{\partial \xi} + J\eta_t \frac{\partial u}{\partial \eta}$$

$$= \left(\frac{\partial (Ju)}{\partial \tau} - u\frac{\partial J}{\partial \tau}\right) + \frac{\partial (J\xi_t u)}{\partial \xi} - u\frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - u\frac{\partial (J\eta_t)}{\partial \eta}$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(u\frac{\partial J}{\partial \tau} + u\frac{\partial (J\xi_t)}{\partial \xi} + u\frac{\partial (J\eta_t)}{\partial \eta}\right)$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(\frac{\partial J}{\partial \tau} + \frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t)}{\partial \eta}\right)u$$

$= 0$ (geometric conservation law)

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi}^{0} + \frac{\partial (J\eta_t u)}{\partial \eta}^{0} = \frac{\partial (Ju)}{\partial \tau}$$

Similar manipulation of term 2, & term 3 (first order spatial derivatives)

term 2:

$$J\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)u^2 = \frac{\partial (J\xi_x u^2)}{\partial \xi} + \frac{\partial (J\eta_x u^2)}{\partial \eta} - u^2\left(\frac{\partial (J\xi_x)}{\partial \xi} + \frac{\partial (J\eta_x)}{\partial \eta}\right)$$

$= 0$ (geometric conservation law)

term 3:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)uv = \frac{\partial (J\xi_y uv)}{\partial \xi} + \frac{\partial (J\eta_y uv)}{\partial \eta} - uv\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\eta_y)}{\partial \eta}\right)$$

$= 0$ (geometric cons. law)

term 4:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right) = J\xi_y \frac{\partial}{\partial \xi}\underbrace{\left(\frac{\partial u}{\partial y}\right)}_{a} + J\eta_y \frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial y}\right)$$

$$= J\xi_y \frac{\partial a}{\partial \xi} + J\eta_y \frac{\partial a}{\partial \eta}$$

$$= \frac{\partial (J\xi_y a)}{\partial \xi} - a\frac{\partial J\xi_y}{\partial \xi} + \frac{\partial (J\eta_y a)}{\partial \eta} - a\frac{\partial (J\eta_y)}{\partial \eta}$$

$$= \frac{\partial (J\xi_y a)}{\partial \xi} + \frac{\partial (J\eta_y a)}{\partial \eta} - a\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\xi_y)}{\partial \eta}\right)$$

$= 0$ (geometric Cons. law)

$$= \frac{\partial}{\partial \xi}\left(J\xi_y \frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial \eta}\left(J\xi_y \frac{\partial u}{\partial y}\right)$$

Therefore, Final strong conservative form:

$$\frac{\partial (Ju)}{\partial \tau} + \left[\frac{\partial}{\partial \xi}(J\xi_x u) + \frac{\partial}{\partial \eta}(J\eta_x u) \right] + \left[\frac{\partial}{\partial \xi}(J\xi_y v) + \frac{\partial}{\partial \eta}(J\eta_y v) \right] = \frac{\partial}{\partial \xi}\left(J\xi_y \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial \eta}\left(J\xi_y \frac{\partial v}{\partial y} \right)$$

$$\boxed{\frac{\partial (Ju)}{\partial \tau} + \frac{\partial}{\partial \xi}\left(J\xi_x u^2 + J\xi_y uv - J\xi_y \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial \eta}\left(J\eta_x u^2 + J\eta_y uv - J\xi_y \frac{\partial u}{\partial y} \right) = 0}$$

(c) Highlighted terms in blue show the 4 geometric conservation laws (in expression above)

Problem 2

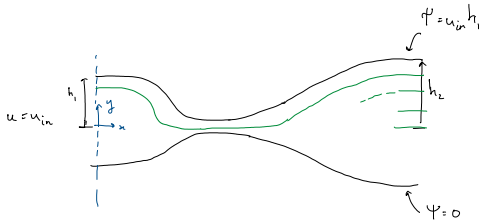
Incompressible potential flow $\nabla^2 \psi = 0$ (ψ: stream function)

(a) Describe the problem

Potential flow is used to model irrotational where the viscosity effects are assumed to be negligible - typically externally flow over solid surfaces or flow at high Reynolds number. In such cases, the thickness of the boundary layer is assumed to be much smaller than the dimen sions of the object.

The given equation is the Laplace equation using stream function. Stream function (ψ) can be defined for any 2D flow such that the stream function is constant along a streamline. Additionally, change in stream function between a pair of streamlines equals the volummetric flowrate between the two streamlines. Finally, lines of constant streamlines are perpendicular to the lines of constant velocity potential (Φ).

(b) Governing equations



$u = \frac{\partial \psi}{\partial y}$ $\quad v = -\frac{\partial \psi}{\partial x}$ 2D, axisymmetric flow ; ψ = stream function (constant along a stream line)

governing equations

$$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\downarrow$$

$$\frac{\partial}{\partial x}(-v) + \frac{\partial u}{\partial y} = 0$$

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0$$

BC's :

(Inflow) $u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$ (outflow) $\left.\frac{\partial \psi}{\partial x}\right|_{x=L} = 0$

(walls) $\psi = 0$ (@ curve 1)

$\psi = u_{in} h_1$ (@ curve 2)

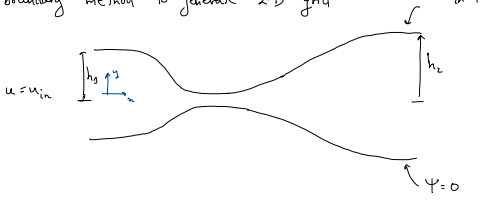
We also use bernoulli's principle to relate pressure & fluid velocity

@ steady state

$$P_{inlet} + \frac{1}{2} \rho u_{in}^2 = \text{constant}$$

@ $x = 0$, $P_{inlet} = P_{atm} = 100{,}000$ Pa

(c) Two-boundary method to generate 2-D grid



$u = \frac{\partial \psi}{\partial y}$ $\quad v = -\frac{\partial \psi}{\partial x}$

$$\int_{(y-h_1)}^{(y+h_1)} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

$$\psi(y) = u_{in}\left((y+h_1) - (y-h_1)\right) = u_{in}(2h_1)$$

$X_1(\xi,\tau) = x(\xi, \eta=0, \tau)$ $\quad X_2(\xi,\tau) = x(\xi, \eta=1, \tau)$

$Y_1(\xi,\tau) = y(\xi, \eta=0, \tau)$ $\quad Y_2(\xi,\tau) = y(\xi, \eta=1, \tau)$

$$\vec{r_1} = X_1 \vec{i} + Y_1 \vec{j}$$

$$\vec{r_2} = X_2 \vec{i} + Y_2 \vec{j}$$

Hermite interpolation :

$$x(\xi,\eta,\tau) = X_1(\xi,\tau) h_1(\eta) + X_2(\xi,\tau) h_2(\eta) + \frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta} h_3(\eta) + \frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta} h_4(\eta)$$

$$y(\xi,\eta,\tau) = Y_1(\xi,\tau) h_1(\eta) + Y_2(\xi,\tau) h_2(\eta)$$

$$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$$

$$y(\xi,\eta,\tau) = Y_1(\xi,\tau)\,h_1(\eta) + Y_2(\xi,\tau)\,h_2(\eta)$$

$$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$$

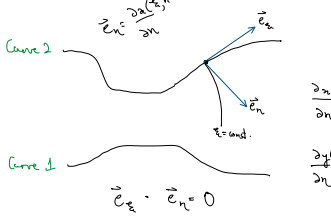
$$h_2(\eta) = -2\eta^3 + 3\eta^2$$

$$h_3(\eta) = \eta^3 - 2\eta^2 + \eta$$

$$h_4(\eta) = \eta^3 - \eta^2$$

$$\vec{e}_\xi = \frac{\partial x_1(\xi,\tau)}{\partial \xi}\vec{i} + \frac{\partial Y_1(\xi,\tau)}{\partial \xi}\vec{j}$$

$$\vec{e}_\eta = \frac{\partial x_2(\xi,\eta=1,\tau)}{\partial \eta}\vec{i} + \frac{\partial y_2(\xi,\eta=1,\tau)}{\partial \eta}\vec{j}$$



$$\vec{e}_\xi \cdot \vec{e}_\eta = 0$$

$$\frac{\partial x_2(\xi,\eta=1,\tau)}{\partial \eta} = -K_2(\xi,\tau)\,\frac{\partial Y_1(\xi,\tau)}{\partial \xi}$$

$$\frac{\partial y_2(\xi,\eta=1,\tau)}{\partial \eta} = +K_2(\xi,\tau)\,\frac{\partial x_1(\xi,\tau)}{\partial \xi}$$

} K-fn. terms

Parametrize the boundaries ; $x = \xi$ $y = f(x) = f(\xi)$

$$\vec{r} = \xi\,\vec{i} + f(\xi)\vec{j}$$

We are given :

Curve 1 : $f_1(x)$

$$\begin{cases} x \in [A,B] \;;\; Y_1 = (D-a_1) - a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_1 = (D-a_2) + a_2 \cos\left[\dfrac{\pi(x-B)}{(C-B)}\right] \end{cases}$$

Curve 2 : $f_2(x)$

$$\begin{cases} x \in [A,B] \;;\; Y_2 = (E+a_1) + a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_2 = (E+a_2) - a_2 \cos\left(\dfrac{\pi(x-B)}{(C-B)}\right) \end{cases}$$

(d) Transform (x,y) to (ξ,η)

$$\nabla^2\psi = 0$$

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 0$$

$$\Rightarrow J\left(\xi_x\frac{\partial}{\partial\xi} + \eta_x\frac{\partial}{\partial\eta}\right)\underbrace{\left(\frac{\partial\psi}{\partial x}\right)}_{f} + J\left(\xi_y\frac{\partial}{\partial\xi} + \eta_y\frac{\partial}{\partial\eta}\right)\underbrace{\left(\frac{\partial\psi}{\partial y}\right)}_{g} = 0$$

} strong conservative (times by J^2)

$$\Rightarrow J\xi_x\frac{\partial f}{\partial\xi} + J\eta_x\frac{\partial f}{\partial\eta} + J\xi_y\frac{\partial g}{\partial\xi} + J\eta_y\frac{\partial g}{\partial\eta} = 0$$

$$\Rightarrow \frac{\partial(J\xi_x f)}{\partial\xi} - f\frac{\partial(J\xi_x)}{\partial\xi} + \frac{\partial(\eta_x J f)}{\partial\eta} - f\frac{\partial(J\eta_x)}{\partial\eta} + \frac{\partial(J\xi_y g)}{\partial\xi} - g\frac{\partial(J\xi_y)}{\partial\xi} + \frac{\partial(J\eta_y g)}{\partial\eta} - g\frac{\partial(J\eta_y)}{\partial\eta} = 0$$

$$\Rightarrow \frac{\partial(J\xi_x f)}{\partial\xi} + \frac{\partial(\eta_x J f)}{\partial\eta} + \frac{\partial(J\xi_y g)}{\partial\xi} + \frac{\partial(J\eta_y g)}{\partial\eta} - f\left(\frac{\partial(J\xi_x)}{\partial\xi} + \frac{\partial(J\eta_x)}{\partial\eta}\right) - g\left(\frac{\partial(J\xi_y)}{\partial\xi} + \frac{\partial(J\eta_y)}{\partial\eta}\right) = 0$$

$$\Rightarrow \boxed{\frac{\partial}{\partial\xi}\left(J\xi_x\frac{\partial\psi}{\partial x} + J\xi_y\frac{\partial\psi}{\partial y}\right) + \frac{\partial}{\partial\eta}\left(J\eta_x\frac{\partial\psi}{\partial x} + J\eta_y\frac{\partial\psi}{\partial y}\right) = 0}$$

$$J\xi_x\left(\xi_{1x}\frac{\partial}{\partial e_1} + \eta_y\frac{\partial}{\partial\eta}\right)\psi = \xi_{1x}\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)$$

$$\therefore \frac{\partial}{\partial\xi}\left(\xi_{1x}\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right) + \xi_y\left(\frac{\partial(J\xi_y\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)\right) + \frac{\partial}{\partial\eta}\left(\eta_x\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_x\psi)}{\partial\eta}\right) + \eta_y\left(\frac{\partial(J\xi_y\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)\right) = 0$$

(e), (f) FDE / FVE's for governing equations + algorithm

$$\left(J^2\xi_x\right)\frac{\partial}{\partial\xi}\left(\xi_x\frac{\partial\psi}{\partial\xi}\right) = \frac{\partial}{\partial\xi}\left[\left(J^2\xi_x^2\right)\frac{\partial\psi}{\partial\xi}\right] - \left(\xi_x\frac{\partial\psi}{\partial\xi}\right)\frac{\partial}{\partial\xi}\left(J^2\xi_x\right)$$

$$\frac{\left(J^2\xi_x^2\right)_{i+\frac12,j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i+\frac12,j} - \left(J^2\xi_x^2\right)_{i-\frac12,j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i-\frac12,j}}{\Delta\xi} - \xi_x\frac{\partial\psi}{\partial\xi}\left[J\frac{\partial}{\partial\xi}(J\xi_x) + \xi_x J\frac{\partial}{\partial\xi}J\right]$$

$$\frac{(J^2\xi_m^2)_{i+\frac{1}{2},j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i+\frac{1}{2},j} - (J^2\xi_x^2)_{i-\frac{1}{2},j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i-\frac{1}{2},j}}{\Delta\xi} \qquad \xi_x\frac{\partial\psi}{\partial\xi}\left[J\frac{\partial}{\partial\xi}(J\xi_x) + \xi_x J\frac{\partial}{\partial\xi}J\right]$$

$$\frac{\psi_{i+1,j}-\psi_{i,j}}{\Delta\xi}$$

$$\frac{\partial}{\partial\xi}(J\xi_x\psi) = \left[(J\xi_x)_{i+\frac{1}{2},j+\frac{1}{2}}\psi_{i+\frac{1}{2},j+\frac{1}{2}} - (Jn)_{i+\frac{1}{2},j-\frac{1}{2}}\psi_{i+\frac{1}{2},j-\frac{1}{2}}\right]/\Delta\xi$$

$$\psi_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\psi_{i+1,j} + \psi_{i,j} + \psi_{i+1,j+1} + \psi_{i,j+1})$$

$$\vec{A}_\xi = (J\xi_x)\vec{i} + (J\xi_y)\vec{j}$$

$$J = x_\xi y_n - y_\xi x_n \quad ; \quad \xi_x = \frac{y_n}{J} \quad ; \quad \xi_y = -\frac{x_n}{J} \quad ; \quad n_x = -\frac{y_\xi}{J} \quad ; \quad n_y = \frac{x_\xi}{J}$$

} Metric coefficients & Jacobian

$$(\xi_x)_{i,j} = \frac{y_{i,j+1} - y_{i,j-1}}{2\Delta n J_{i,j}} \qquad (\xi_y)_{i,j} = -\frac{x_{i,j+1} - x_{i,j-1}}{2\Delta n J_{i,j}}$$

$$(n_x)_{i,j} = -\frac{y_{i+1,j} - y_{i-1,j}}{2\Delta\xi J_{i,j}} \qquad (n_y)_{i,j} = \frac{x_{i+1,j} - x_{i-1,j}}{2\Delta\xi J_{i,j}}$$

$$J_{i,j} = \left(\frac{x_{i+1,j} - x_{i-1,j}}{2\Delta\xi}\right)\left(\frac{y_{i,j+1} - y_{i,j-1}}{2\Delta n}\right) - \left(\frac{y_{i+1,j} - y_{i-1,j}}{2\Delta\xi}\right)\left(\frac{x_{i,j+1} - x_{i,j-1}}{2\Delta n}\right)$$

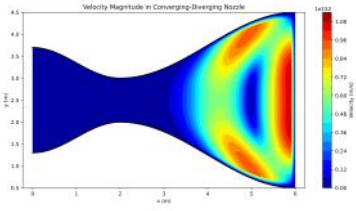
Algorithm:

1) Generate grid via Two-boundary method
   - Use IL, JL = 41 (from lecture) → step sizes of $\Delta\xi = \frac{1}{(IL-1)}$; $\Delta n = \frac{1}{(JL-1)}$
   - Use hermite interpolation
     - → (parameterized curves, computed basis functions, using K factor of 0.08)
     - → Gives grid for X, Y
2) Set initial conditions (u_{in}, P_{in}, ρ)
3) Compute Jacobian & metric coefficients for transformation
4) Finite difference loops for solving for ψ in the transformed domain
5) Inverse transform & using u & v relation to obtain velocity in the physical domain
6) Plotting contour plot

(g), (h) Code + solutions for velocity = 100 m/s ; $P = 100{,}000\ P_a$; $\rho = 1000\ kg/m^3$

See attached document for code
(debugging via LLM software)

The Two-boundary method code was borrowed from the fortran code that was shared in lecture 16, slide 35.



(i) Discuss accuracy of results.

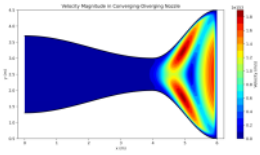
There seems to be a bug in the code which prohibited a physical solution that can be analyzed; however, we can still discuss about the results we expect from a solution to the converging-diverging nozzle

Due the converging geometry, we expect that the velocity will be the highest in the constriction of the nozzle (i.e. for this geometry it should be at x=2); as the fluid moves past the constriction into the diverging part of the nozzle. Since we are assuming incompressible potential flow, the diverging section would decelerate the flow (as the area increases, and velocity decreases). However, this contradicts real compressible flows where you could have supersonic acceleration.

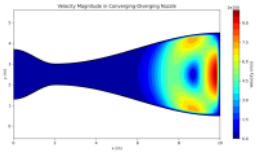
Extra credit: Changing the geometry of the nozzle

Moving the value of B to a higher value of x i.e. B = 4, should simulate the nozzle contraction moving to a later value and hence the expected velocity peak also changes around this constriction. Changing the C parameter will just extend the diverging part of the nozzle which will pretty much make the flow to be converted to flow in a tube (i.e. the peak velocities should decelerate pretty quickly - my code does not show this effect accurately sorry)

Something that would have been also interesting to look at (if my code would have worked) is changing the geometry angles which is known to minimize flow separation and have smooth acceleration to the throat. The throat length can also be varied to stabilize flow before divergence.



Varying the B parameter



Changing the E parameter

HW6

Tuesday, 8 April 2025 9:47 AM

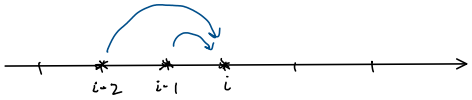
Problem 1

(i) Derive difference operator for $\left(\frac{dT}{dx}\right)_{I_L}$ w/ $O(\Delta x^2)$ accuracy

(1) No. of terms to be kept in Taylor series

N = order of derivative + order of accuracy
= 1 + 2
= 3

(2) Stencil



(3) N-1 truncated Taylor series about T_{I_L}

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{\Delta x^2}{2!} - O(\Delta x^3)$$

$$T_{I_L-2} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}(2\Delta x) + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{(2\Delta x)^2}{2!} - O(\Delta x^3)$$

(4) Solving coupled linear equations

Subtracting the two terms:

$$T_{I_L-1} - T_{I_L-2} = \left(\frac{dT}{dx}\right)_{I_L}\Delta x - \left(\frac{3}{2}\Delta x^2\right)\left(\frac{d^2T}{dx^2}\right)_{I_L}$$

$$\left(\frac{d^2T}{dx^2}\right)_{I_L} = \left[(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\Delta x\right]\frac{2}{3(\Delta x^2)}$$

Substituting here

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left(\frac{d^2T}{dx^2}\right)_{I_L}\frac{\Delta x^2}{2!} - O(\Delta x^3)$$

$$\therefore\ T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \left[(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\Delta x\right]\frac{2}{3\Delta x^2}\cdot\frac{\Delta x^2}{2}$$

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \frac{1}{3}(-T_{I_L-1} + T_{I_L-2}) + \left(\frac{dT}{dx}\right)_{I_L}\left(\frac{1}{3}\Delta x\right)$$

$$T_{I_L-1} = T_{I_L} - \left(\frac{dT}{dx}\right)_{I_L}\Delta x + \frac{1}{3}\left(-T_{I_L-1} + T_{I_L-2}\right) + \left(\frac{dT}{dx}\right)_{I_L}\left(\frac{1}{3}\Delta x\right)$$

$$\left(\frac{dT}{dx}\right)_{I_L}\left(\Delta x - \frac{1}{3}\Delta x\right) = T_{I_L} - T_{I_L-1} - \frac{1}{3}T_{I_L-1} + \frac{1}{3}T_{I_L-2}$$

$$\left(\frac{dT}{dx}\right)_{I_L}\left(\frac{2}{3}\Delta x\right) = T_{I_L} - \frac{4}{3}T_{I_L-1} + \frac{1}{3}T_{I_L-2}$$

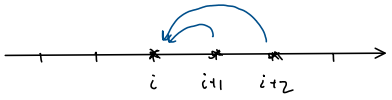
$$\left(\frac{dT}{dx}\right)_{I_L} = \left(\frac{3}{2\Delta x}\right)\left[\frac{3T_{I_L} - 4T_{I_L-1} + T_{I_L-2}}{3}\right]$$

$$\boxed{\left(\frac{dT}{dx}\right)_{I_L} = \frac{3T_{I_L} - 4T_{I_L-1} + T_{I_L-2}}{2\Delta x} + O(\Delta x^2)}$$

(ii) Following a similar strategy for calculating difference operator for $\left(\frac{dT}{dx}\right)_1$

(1) $N = 3$

(2)



(3) $$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(\Delta x)^2}{2} + O(\Delta x^3)$$

$$T_3 = T_1 + \left(\frac{dT}{dx}\right)_1 2\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(2\Delta x)^2}{2} + O(\Delta x^3)$$

(4) Subtracting the equations :

$$T_2 - T_3 = -\left(\frac{dT}{dx}\right)_1\Delta x - \left(\frac{d^2T}{dx^2}\right)_1\left(\frac{3}{2}\Delta x^2\right)$$

$$\left(\frac{d^2T}{dx^2}\right)_1 = \left[(T_3 - T_2) - \left(\frac{dT}{dx}\right)_1\Delta x\right]\frac{2}{3\Delta x^2}$$

Substituting

$$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left(\frac{d^2T}{dx^2}\right)_1\frac{(\Delta x)^2}{2} + O(\Delta x^3)$$

$$T_2 = T_1 + \left(\frac{dT}{dx}\right)_1\Delta x + \left[(T_3 - T_2) - \left(\frac{dT}{dx}\right)_1\Delta x\right]\frac{\Delta x^2}{2}\cdot\frac{2}{3\Delta x^2}$$

$$\left(\frac{dT}{dx}\right)_1\left(-\frac{2\Delta x}{3}\right) = T_1 - T_2 + \frac{T_3}{3} - \frac{T_2}{3}$$

$$= \left[\frac{3T_1 - 4T_2 + T_3}{3}\right]\frac{3}{(-2\Delta x)}$$

$$\left(\frac{dT}{dx}\right)_1 = \frac{-3T_1 + 4T_2 - T_3}{2\Delta x} + O(\Delta x^2)$$

Problem 2

4th order compact operator

$$c\frac{\partial U}{\partial x} = \nu\frac{\partial^2 U}{\partial x^2} \quad ; \quad @\ x=0,\ U=1 \quad @\ x=L=10,\ \frac{dU}{dx}=2$$

$$\downarrow$$

$$cF_i - \nu S_i = 0 \qquad \frac{cL}{\nu} = 2$$

$\left(\frac{\partial U}{\partial x}\right)_i$ $\left(\frac{\partial^2 U}{\partial x^2}\right)_i$

Compact differencing yields:

$$F_{i+1} + 4F_i + F_{i-1} = \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) \quad \cdots (1)$$

Derived via operator approach

$$S_{i+1} + 10S_i + S_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1}) \quad \cdots (2)$$

We know: $cF_i = \nu S_i$

$S_i = \frac{c}{\nu}F_i$ → substituting in above equation

$$\therefore \quad \frac{c}{\nu}F_{i+1} + \frac{10c}{\nu}F_i + \frac{c}{\nu}F_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1}) \quad \cdots (3)$$

Similarly, $F_i = \frac{\nu}{c}S_i$

$$\frac{\nu}{c}S_{i+1} + 4S_i + S_{i-1} = \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) \quad \cdots (4)$$

We can combine (1) (3) in matrix form

From (1)

$$F_{i+1} + 4F_i + F_{i-1} - \frac{3}{\Delta x}(U_{i+1} - U_{i-1}) = 0$$

$$\Rightarrow \left[F_{i+1} - \frac{3}{\Delta x}U_{i+1}\right] + 4F_i + \left[\frac{3}{\Delta x}U_{i-1} + F_{i-1}\right] = 0$$

From (3)

$$\frac{c}{\nu}F_{i+1} + \frac{10c}{\nu}F_i + \frac{c}{\nu}F_{i-1} = \frac{12}{\Delta x^2}(U_{i+1} - 2U_i + U_{i-1})$$

$$= \left[\frac{c}{\nu} F_{i+1} - \frac{12}{\Delta x^2} U_{i+1}\right] + \left[10\frac{c}{\nu} F_i + \frac{24}{\Delta x^2} U_i\right] + \left[\frac{c}{\nu} F_{i-1} - \frac{12}{\Delta x^2} U_{i-1}\right] = 0$$

Combining above expressions in matrix form:

$$\underbrace{\begin{bmatrix} -\frac{3}{\Delta x} & 1 \\ -\frac{12}{\Delta x^2} & c/\nu \end{bmatrix}}_{L_i} \begin{bmatrix} U_{i+1} \\ F_{i+1} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 4 \\ \frac{24}{\Delta x^2} & 10\frac{c}{\nu} \end{bmatrix}}_{M_i} \begin{bmatrix} U_i \\ F_i \end{bmatrix} + \underbrace{\begin{bmatrix} 3/\Delta x & 1 \\ -12/\Delta x^2 & c/\nu \end{bmatrix}}_{N_i} \begin{bmatrix} U_{i-1} \\ F_{i-1} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{d_i}$$

If we define $V_i = \begin{bmatrix} U_i \\ F_i \end{bmatrix}$; where $V_{i-1} = \begin{bmatrix} U_{i-1} \\ F_{i-1} \end{bmatrix}$ we obtain:

$$\boxed{L_i V_{i-1} + M_i V_i + N_i V_{i+1} = d_i} \quad \cdots (5)$$

↳ Solving this system of equations will yield $O(\Delta x^4)$ accurat solution

Apply Boundary conditions

Interior points

$L_i V_{i-1} + M_i V_i + N_i V_{i+1} = d_i$

↳ One-sided differencing that relates $\frac{dU}{dx}$ at the boundary to two neighboring points

@ $x = 0$ $(i=1)$ $U = 1$ $\quad V_1 = \begin{bmatrix} 1 \\ F_1 \end{bmatrix}$

$L_2 V_1 + M_2 V_2 + N_2 V_3 = d_1$

@ $x = L$ $\frac{dU}{dx} = F = 2$ $\quad V_{IL} = \begin{bmatrix} U_{IL} \\ 2 \end{bmatrix}$

$L_{IL-1} V_{IL-2} + M_{IL-1} V_{IL-1} + N_{IL} V_{IL} = d$

Overall algorithm

1. Discretize the domain ($x_i = (i-1)\Delta x$; $i = 1, 2, 3 \cdots n, n+1$)
   ↳ grid is initialized for given Δx
2. Use compact differencing to discretize ODE (arriving at eq (5) from above)
3. Apply given boundary conditions & solve for interior points
   → Iterate using for loop
4. Plot and analyze results

HW7

Wednesday, 23 April 2025 9:41 AM

Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial (x+y)u}{\partial y} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x \partial y}\right) + au^2$$

$$0 \le x \le L_x \qquad 0 \le y \le L_y$$

(a) FDE for interior points

Domain discretized as such: $x_i = (i-1)\Delta x$, $i = 1, 2, 3 \cdots, IL$

$y_j = (j-1)\Delta y$, $j = 1, 2, 3 \cdots, JL$

Let $u^n_{i,j}$ represent value of u at (i,j) spatial point & n time step

i) Self-starting second-order implicit for time differencing:

↳ Crank-Nicolson scheme : $\frac{\partial u}{\partial t} \approx \frac{u^{n+1}_{i,j} - u^n_{i,j}}{\Delta t} = \frac{1}{2}$

(ii) Second-order central differencing for spatial derivatives:

$$\frac{\partial u^2}{\partial x} = \frac{(u^n_{i+1,j})^2 - (u^n_{i-1,j})^2}{2\Delta x}$$

$$\frac{\partial (x+y)u}{\partial y} = \frac{(x_i + y_{j+1})u^n_{i,j+1} - (x_i + y_{j-1})u^n_{i,j-1}}{2\Delta y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}}{(\Delta x)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1}}{(\Delta y)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{u^n_{i+1,j+1} - u^n_{i+1,j-1} - u^n_{i-1,j+1} + u^n_{i-1,j-1}}{4\Delta x \Delta y}$$

(iii) $\frac{\partial u}{\partial t} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x \partial y}\right) + au^2 - \frac{\partial u^2}{\partial x} - \frac{\partial (x+y)u}{\partial y}$

$$\frac{u^{n+1}_{i,j} - u^n_{i,j}}{\Delta t} = \left[\nu\left(D_x^2 + D_y^2 + D_x D_y\right)u_{i,j} + \underbrace{a(u_{i,j})^2}_{\text{source term}} - D_x(u_{i,j})^2 - D_y(x_i + y_j)u_{i,j}\right]^{n+1}$$

⇒ Linearizing and writing in operator form:

$$(u^2_{i,j})^{n+1} \approx (u^2_{ij})^n + 2u^n_{ij}\underbrace{(u^{n+1} - u^n)}_{\Delta u^{n+1}_{ij}} + \mathcal{O}(\Delta t^2)$$

$$u^{n+1}_{ij} = (u^{n+1} - u^n) + u^n = \Delta u^{n+1}_{ij} + u^n_{ij}$$

$$\therefore \frac{\Delta u^{n+1}_{ij}}{\Delta t} = \left[\nu(D_x^2 + D_y^2 + D_x D_y)\underbrace{u^{n+1}_{ij}}_{\Delta u^{n+1}_{ij} + u^n_{ij}} + a\left((u^2_{ij})^n + 2u^n_{ij}\Delta u^{n+1}_{ij}\right) - D_x\left((u^2_{ij})^n + 2u^n_{ij}\Delta u^{n+1}_{ij}\right) - D_y(x_i+y_j)u^{n+1}_{ij}\right]$$

In operator form:

$$\Delta u^{n+1} - \Delta t\left[\nu(D_x^2 + D_y^2 + D_x D_y)\Delta u^{n+1} - a(2u^n \Delta u^{n+1}) + D_x(2u^n \Delta u^{n+1}) + D_y(x+y)\Delta u^{n+1}\right] = \left[\nu(D_x^2 + D_y^2 + D_z^2)u^n + a(u^2)^n - D_x(u^2)^n - D_y(x+y)u^n\right]\Delta t$$

We can treat the cross-derivative term explicitly (while maintaining second-order accuracy)

↳ $D_x D_y \Delta u^{n+1} \approx D_x D_y(u^n)$ → this term is then moves to the RHS

⇒ $$1 - \Delta t\left[\nu(D_x^2 + D_y^2) - a(2u^n) + D_x(2u^n) + D_y(x+y)\right]\Delta u^{n+1} = \overbrace{\left[\nu(D_x^2 + D_y^2 + D_z^2)u^n + a(u^2)^n - D_x(u^2)^n - D_y(x+y)u^n - D_x D_y u^n\right]\Delta t}^{\text{RHS}}$$

$$1 - \Delta t\left[\nu(D_x^2 + D_y^2) - a(2u^n) + D_x(2u^n) + D_y(x+y)\right]\Delta u^{n+1} = \text{RHS}$$

Now we can factor approximately (2 spatial dimensions), split the x & y factors & the apply the split operators at grid points ij & replace spatial derivatives by difference operators (these source term is put in w/ x factor — although this is not the best method)

$$\underbrace{\left\{1 - \Delta t\left[\nu D_x^2 - a2u^n + D_x(2u^n)\right]\right\}\left\{1 + \Delta t\left(\nu D_y^2 + D_y(x+y)\right)\right\}}_{\Delta u^*}\Delta u^{n+1} = \text{RHS}$$

⇒ $$\Delta u^*_{i,j} + \Delta t D_x^2 \Delta u^*_{i,j} - \Delta t a 2[u^n \Delta u^*]_{i,j} + \Delta t D_x(2u^n \Delta u^*)_{i,j} = \text{RHS}_{i,j}$$

$$\Delta u^*_{i,j} + \Delta t\left(\frac{\Delta u^*_{i+1,j} - 2\Delta u^*_{i,j} + \Delta u^*_{i-1,j}}{\Delta x^2}\right) - \Delta t a 2[u^n \Delta u^*]_{i,j} + \Delta t\left(\frac{(2u^n \Delta u^*)_{i+1,j} - (2u^n \Delta u^*)_{i-1,j}}{2\Delta x}\right) = \text{RHS}_{i,j}$$

$$L_{i,j}\,\Delta u^{*}_{i-1,j} + M_{i,j}\,\Delta u^{*}_{i,j} + N_{i,j}\,\Delta u^{*}_{i+1,j} = RHS_{i,j} \longrightarrow \text{Eq } (1)$$

$\left(\frac{\Delta t}{\Delta x^2} - \frac{2u^{n}_{i-1,j}\Delta t}{2\Delta x}\right)$ $\left(1 - \frac{2\Delta t}{\Delta x^2} - \Delta t\, a\, 2u^{n}_{i,j}\right)$ $\left(\frac{\Delta t}{\Delta x^2} + \frac{\Delta t\, 2u^{n}_{i+1,j}}{2\Delta x}\right)$ $\quad j = 1, 2, 3 \ldots JL-1$

Therefore, our tridiagonal matrix for interior points:

$$\begin{bmatrix} M_{2,j} & N_{2,j} & & & \\ L_{3,j} & M_{3,j} & N_{3,j} & & \\ & \ddots & \ddots & \ddots & \\ & & L_{JL-2,j} & M_{JL-2,j} & \\ & & & & M_{JL-1,j} \end{bmatrix} \begin{bmatrix} \Delta u^{*}_{2,j} \\ \Delta u^{*}_{3,j} \\ \vdots \\ \Delta u^{*}_{JL-1,j} \end{bmatrix} = \begin{bmatrix} RHS_{2,j} \\ RHS_{3,j} \\ \vdots \\ RHS_{JL-1,j} \end{bmatrix}$$

↳ Boundary conditions need to be handled separately (above solution only for interior points)

Similarly,

$$\Delta u^{*} = \left\{1 + \Delta t\,(\nu D_y^2 + D_y(v+y))\right\}\Delta u^{n+1}$$

$$= \left\{1 + \Delta t\,\nu\left(\frac{\Delta u^{n+1}_{i,j+1} - 2\,\Delta u^{n+1}_{i,j} + \Delta u^{n+1}_{i,j-1}}{\Delta y^2}\right) + \Delta t\left(\frac{(v_{i,j+1}+y_j)\,\Delta u^{n+1}_{i,j+1} - (v_{i,j-1}+y_j)\,\Delta u^{n+1}_{i,j-1}}{2\Delta y}\right)\right.$$

$$\Delta u^{*} = L_{i,j}\,\Delta u^{n+1}_{i,j-1} + M_{i,j}\,\Delta u^{n+1}_{i,j} + N_{i,j}\,\Delta u^{n+1}_{i,j+1} \longrightarrow \text{Eq } (2)$$

$\left(\frac{\Delta t\,\nu}{\Delta y^2} - \frac{(v_{i,j}+y_j)\Delta t}{2\Delta y}\right)$ $\left(1 - \frac{2\Delta t\,\nu}{\Delta y^2}\right)$ $\left(\frac{\Delta t\,\nu}{\Delta y^2} + \frac{\Delta t\,(v_{i,j}+y_j)}{2\Delta y}\right)$

A similar tridiagonal matrix can be constructed for Eq (2) for $i = 2, 3 \ldots IL-1$ which are the interior points

$$\begin{bmatrix} M_{i,2} & N_{i,2} & & & \\ L_{i,3} & M_{i,3} & N_{i,3} & & \\ & \ddots & \ddots & \ddots & \\ & & L_{i,JL-1} & M_{i,JL-2} & N_{i,JL-2} \\ & & & & M_{i,JL-1} \end{bmatrix} \begin{bmatrix} \Delta u^{n+1}_{i,2} \\ \Delta u^{n+1}_{i,3} \\ \vdots \\ \Delta u^{n+1}_{i,JL-1} \end{bmatrix} = \begin{bmatrix} \Delta u^{*}_{i,2} \\ \Delta u^{*}_{i,3} \\ \vdots \\ \Delta u^{*}_{i,JL-1} \end{bmatrix}$$

(b) Solution algorithm

1) Using Eq. (1) to solve for $\Delta u^{*}_{i,j}$ from $j = 2, 3 \ldots JL-1$ via Thomas algorithm
2) Use Eq (2) to solve for $\Delta u^{n+1}_{i,j}$ (using constructed tridiagonal matrix) from $i = 2, 3, \ldots IL-1$ using Thomas algorithm
3) Calculate $u^{n+1}_{i,j} = u^{n}_{i,j} + \Delta u^{n+1}_{i,j}$

↳ Repeat above steps until convergence of duration of interest.

Problem 2

Compressible, inviscid flow; ideal gas; 2D converging-diverging nozzle

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

(a) Describe the problem; given where the Mach number and the static temperature at the duct inlet are 0.2 and 300 K, respectively and the pressure at the duct exit is 1 atm.

The given governing equation is the continuity equation for inviscid flow, where variables U, F, G are related to the flow quantities such as mass/density, velocity, pressure and energy. The 2D converging-diverging nozzle has only one inlet (left boundary) and outlet (right boundary), with no flow through the top and bottom walls

The Mach number represents the ratio of the flow velocity to the speed of sound in the medium. Since the Mach number at the inlet is 0.2, the flow is subsonic at the inlet.

(b) Formulate the governing equations for the problem by using the Euler equations with appropriate boundary conditions at the walls and the inflow and outflow boundaries.

The Euler equations for compressible, inviscid flow are given by:

1) Continuity eq.: $\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0$

2) Momentum eq.: $\frac{\partial(\rho u)}{\partial t} + \frac{\partial}{\partial x}(\rho u^2 + P) + \frac{\partial}{\partial y}(\rho u v) = 0$

$\frac{\partial(\rho v)}{\partial t} + \frac{\partial}{\partial x}(\rho u v) + \frac{\partial}{\partial y}(\rho v^2 + P) = 0$

Combining these equations yields the given expression

$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial}{\partial x}(\rho u v) + \frac{\partial}{\partial y}(\rho v^2 + P) = 0$$

3) Energy equation : $\frac{\partial e}{\partial t} + \frac{\partial}{\partial x}(u(e+P)) + \frac{\partial}{\partial y}(v(e+P)) = 0$

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

where $P = (\gamma - 1)\left(e - \frac{u^2+v^2}{2\rho}\right)$

↓ is given by Bernoulli's principle

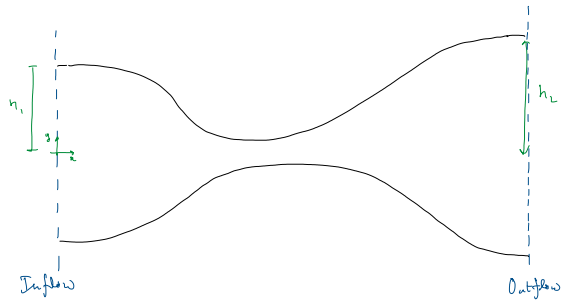
Boundary conditions :

@ walls : $u = v = 0$

@ outflow : $P = 1$ atm

@ inflow : $M = 0.2$, $T = 300$ K

For an ideal gas, {P,V,T} relationship is an additional equation that defines the variables.



(c) Generate a boundary-fitted grid for the 2-D nozzle geometry by using the two-boundary method, where grid lines should intersect boundary curves orthogonally.

$X_1(\xi,\tau) = x(\xi, \eta=0, \tau)$ $\quad X_2(\xi,\tau) = x(\xi, \eta=1, \tau)$

$Y_1(\xi,\tau) = y(\xi, \eta=0, \tau)$ $\quad Y_2(\xi,\tau) = y(\xi, \eta=1, \tau)$

$$\vec{r}_1 = X_1\vec{i} + Y_1\vec{j}$$

$$\vec{r}_L = X_2\vec{i} + Y_2\vec{j}$$

Hermite interpolation : $x(\xi,\eta,\tau) = X_1(\xi,\tau)h_1(\eta) + X_2(\xi,\tau)h_2(\eta) + \frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta}h_3(\eta) + \frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta}h_4(\eta)$

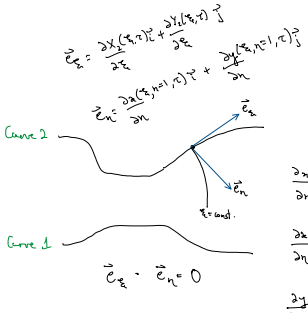
$y(\xi,\eta,\tau) = Y_1(\xi,\tau)h_1(\eta) + Y_2(\xi,\tau)h_2(\eta)$

$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$

$h_2(\eta) = -2\eta^3 + 3\eta^2$

$h_3(\eta) = \eta^3 - 2\eta^2 + \eta$

$h_4(\eta) = \eta^3 - \eta^2$



$$\frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta} = -K_2(\xi,\tau)\frac{\partial Y_2(\xi,\tau)}{\partial \xi}$$

$$\frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta} = +K_1(\xi,\tau)\frac{\partial Y_1(\xi,\tau)}{\partial \xi}$$

} K - functions

$$\frac{\partial y}{\partial \eta}(\xi,\eta=1,\tau) = \frac{\partial X_2}{\partial \xi} = \frac{C}{I_L}$$

$$\frac{\partial y}{\partial \eta}(\xi,\eta=0,\tau) = \frac{\partial X_1}{\partial \xi} = \frac{C}{I_L}$$

Parametrize the boundaries ; $x = \xi$ $\quad y = f(\lambda) = f(\xi)$

$$\vec{r} = \xi\,\vec{i} + f(\xi)\vec{j}$$

We are given :

Curves 1 $f_1(x)$:

$$\begin{cases} x \in [A,B] \;;\; Y_1 = (D-a_1) - a_1 \cos\left[\frac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_1 = (D-a_2) + a_2 \cos\left[\frac{\pi(x-B)}{(C-B)}\right] \end{cases}$$

Curve 2 : $f_2(x)$

$$\begin{cases} x \in [A,B] \;;\; y_2 = (E+a_1) + a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; y_2 = (E+a_2) - a_2 \cos\left(\dfrac{\pi(x-B)}{(C-B)}\right) \end{cases}$$

$$y(\xi,\eta) = y_1\left(x(\xi)\right) + \eta\left(y_2(x(\xi)) - y_1(x(\xi))\right)$$

(d) Transform the governing equations and boundary conditions to boundary-fitted generalized coordinates.

$$(t,x,y) \rightarrow (\tau,\xi,\eta) \quad ; \quad \begin{aligned} \tau &= \tau(t,x,y) \\ \eta &= \eta(t,x,y) \\ \xi &= \xi(t,x,y) \end{aligned}$$

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

$$\frac{\partial}{\partial t} = \underbrace{\frac{\partial \tau}{\partial t}}_{\tau_t}\frac{\partial}{\partial \tau} + \underbrace{\frac{\partial \eta}{\partial t}}_{\eta_t}\frac{\partial}{\partial \eta} + \underbrace{\frac{\partial \xi}{\partial t}}_{\xi_t}\frac{\partial}{\partial \xi}$$

$$\frac{\partial}{\partial x} = \frac{\partial \tau}{\partial x}\frac{\partial}{\partial \tau} + \frac{\partial \eta}{\partial x}\frac{\partial}{\partial \eta} + \frac{\partial \xi}{\partial x}\frac{\partial}{\partial \xi}$$

$$\frac{\partial}{\partial y} = \frac{\partial \tau}{\partial y}\frac{\partial}{\partial \tau} + \frac{\partial \eta}{\partial y}\frac{\partial}{\partial \eta} + \frac{\partial \xi}{\partial y}\frac{\partial}{\partial \xi}$$

$$\left(\tau_t\frac{\partial}{\partial \tau} + \eta_t\frac{\partial}{\partial \eta} + \xi_t\frac{\partial}{\partial \xi}\right)U + \left(\tau_x\frac{\partial}{\partial \tau} + \eta_x\frac{\partial}{\partial \eta} + \xi_x\frac{\partial}{\partial \xi}\right)F + \left(\tau_y\frac{\partial}{\partial \tau} + \eta_y\frac{\partial}{\partial \eta} + \xi_y\frac{\partial}{\partial \xi}\right)G = 0$$

↳ Transforming and converting to strong conservation form (multiply by J)

$$\Rightarrow \left[\frac{\partial}{\partial \tau}(J\tau_t U) - U\frac{\partial}{\partial \tau}(J\tau_t) + \frac{\partial}{\partial \eta}(J\eta_t U) - U\frac{\partial}{\partial \eta}(J\eta_t) + \frac{\partial}{\partial \xi}(J\xi_t U) - U\frac{\partial}{\partial \xi}(J\xi_t)\right]$$
$$+\left[\frac{\partial}{\partial \tau}(J\tau_x F) - F\frac{\partial}{\partial \tau}(J\tau_x) + \frac{\partial}{\partial \eta}(J\eta_x F) - F\frac{\partial}{\partial \eta}(J\eta_x) + \frac{\partial}{\partial \xi}(J\xi_x F) - F\frac{\partial}{\partial \xi}(J\xi_x)\right]$$
$$+\left[\frac{\partial}{\partial \tau}(\tau_y G J) - G\frac{\partial}{\partial \tau}(\tau_y J) + \frac{\partial}{\partial \eta}(\eta_y J G) - G\frac{\partial}{\partial \eta}(\eta_y J) + \frac{\partial}{\partial \xi}(\xi_y J G) - G\frac{\partial}{\partial \xi}(\xi_y J)\right] = 0$$

$$\Rightarrow \left[\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \eta}(J\eta_t U) + \frac{\partial}{\partial \xi}(J\xi_t U)\right] + \left[\frac{\partial}{\partial \tau}(J\tau_x F) + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\frac{\partial}{\partial \tau}(\tau_y G J) + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right]$$
$$-U\left[\frac{\partial}{\partial \tau}(J\tau_t) + \frac{\partial}{\partial \eta}(J\eta_t) + \frac{\partial}{\partial \xi}(J\xi_t)\right] - F\left[\frac{\partial}{\partial \tau}(J\tau_x) + \frac{\partial}{\partial \eta}(J\eta_x) + \frac{\partial}{\partial \xi}(J\xi_x)\right] - G\left[\frac{\partial}{\partial \tau}(J\tau_y) + \frac{\partial}{\partial \eta}(\eta_y J) + \frac{\partial}{\partial \xi}(\xi_y J)\right] = 0$$

If we assume $t = \tau$ & $\tau \neq f(x,y)$ & use geometric conservation laws
↳ Then our transformed governing equation (in strong conservative form) becomes:

$$\left[\frac{\partial}{\partial \tau}(J\tau_t U) + \cancel{\frac{\partial}{\partial \eta}(J\eta_t U)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_t U)}^{\,0}\right] + \left[\cancel{\frac{\partial}{\partial \tau}(J\tau_x F)}^{\,0} + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\cancel{\frac{\partial}{\partial \tau}(\tau_y G J)}^{\,0} + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right]$$
$$-U\left[\frac{\partial}{\partial \tau}(J\tau_t) + \cancel{\frac{\partial}{\partial \eta}(J\eta_t)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_t)}^{\,0}\right] - F\left[\frac{\partial}{\partial \tau}(J\tau_x) + \cancel{\frac{\partial}{\partial \eta}(J\eta_x)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_x)}^{\,0}\right] - G\left[\frac{\partial}{\partial \tau}(J\tau_y) + \cancel{\frac{\partial}{\partial \eta}(\eta_y J)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(\xi_y J)}^{\,0}\right] = 0$$

$$\left[\frac{\partial}{\partial \tau}(J\tau_t U)\right] + \left[\frac{\partial}{\partial \tau}(J\tau_x F) + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\frac{\partial}{\partial \tau}(\tau_y G J) + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right] = 0$$

$$\therefore \boxed{\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \xi}(J\xi_x F + J\xi_y G) + \frac{\partial}{\partial \eta}(J\eta_x F + J\eta_y G) = 0}$$

$$\tilde{U} = \xi_x u + \xi_y v$$
$$\tilde{V} = \eta_x u + \eta_y v$$

Transforming the boundary conditions

inlet wall : $\xi = 0$ plane

outlet wall : $\xi = 1$ plane

bottom wall : $\eta = 0$ } Zero mass flux through walls $\quad \tilde{U}n_x - \tilde{V}\xi_x = 0$

top wall : $\eta = 1$ }

(e) Derive the FDE/FVEs for the governing equations by using the implicit Euler method with first-order upwinding using flux-vector splitting (any variation) that maintains the conservative property.

Flux-splitting scheme

$$\frac{\partial F}{\partial U} = \begin{bmatrix} \frac{\partial F_1}{\partial \rho} & \frac{\partial F_1}{\partial m} & \frac{\partial F_1}{\partial e} & \frac{\partial F_1}{\partial n} \\ \frac{\partial F_2}{\partial \rho} & \frac{\partial F_2}{\partial m} & \frac{\partial F_2}{\partial e} & \frac{\partial F_2}{\partial n} \\ \frac{\partial F_3}{\partial \rho} & \frac{\partial F_3}{\partial m} & \frac{\partial F_3}{\partial e} & \frac{\partial F_3}{\partial n} \\ \frac{\partial F_4}{\partial \rho} & \frac{\partial F_4}{\partial m} & \frac{\partial F_4}{\partial e} & \frac{\partial F_4}{\partial n} \end{bmatrix}$$

$$F = \begin{bmatrix} m \\ \frac{m^2}{\rho} + \gamma e - e - \frac{\gamma}{2\rho}(m^2+n^2) + \frac{(m^2+n^2)}{2\rho} \\ mn/\rho \\ \frac{e\rho}{\rho} + \frac{m}{\rho}\gamma e - \frac{\rho\gamma e}{\rho} - \frac{m}{2\rho^2}\gamma(m^2+n^2) + \frac{m}{2\rho^2}(m^2+n^2) \end{bmatrix}$$

$$\frac{\partial G}{\partial U} = \begin{bmatrix} \frac{\partial G_1}{\partial \rho} & \frac{\partial G_1}{\partial m} & \frac{\partial G_1}{\partial e} & \frac{\partial G_1}{\partial n} \\ \frac{\partial G_2}{\partial \rho} & \frac{\partial G_2}{\partial m} & \frac{\partial G_2}{\partial e} & \frac{\partial G_2}{\partial n} \\ \frac{\partial G_3}{\partial \rho} & \frac{\partial G_3}{\partial m} & \frac{\partial G_3}{\partial e} & \frac{\partial G_3}{\partial n} \\ \frac{\partial G_4}{\partial \rho} & \frac{\partial G_4}{\partial m} & \frac{\partial G_4}{\partial e} & \frac{\partial G_4}{\partial n} \end{bmatrix}$$

$$G = \begin{bmatrix} n \\ mn/\rho \\ n^2/\rho + \gamma e - e - \gamma\left(\frac{m^2+n^2}{2\rho}\right) + \frac{(m^2+n^2)}{2\rho} \\ \frac{e\rho}{\rho} + \frac{n}{\rho}\gamma e - \frac{\rho\gamma e}{\rho} - \frac{\gamma}{\rho}\left(\frac{m^2+n^2}{2\rho}\right) + \frac{n}{2\rho^2}(m^2+n^2) \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{m^2}{\rho^2} + \frac{\gamma}{2}(m^2+n^2) - \frac{(m^2+n^2)}{\rho^2} & \frac{2m}{\rho} - \frac{\gamma m}{\rho} + \frac{m}{\rho} & (\gamma-1) & \frac{\gamma n}{\rho} + \frac{n}{\rho} \\ -\frac{mn}{\rho^2} & \frac{n}{\rho} & 0 & \frac{m}{\rho} \\ \left(-\frac{\gamma e}{\rho^2} + \frac{(n\gamma - m)(m^2+n^2)}{\rho^2}\right) & \left(-\frac{\gamma e}{\rho} - \frac{3\gamma m^2}{2\rho^2} + \frac{3m^2}{2\rho}\right) & \frac{m\gamma}{\rho} & \frac{\gamma mn}{\rho^2} + \frac{mn}{\rho^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ -\frac{mn}{\rho^2} & \frac{n}{\rho} & 0 & \frac{m}{\rho} \\ \left(-\frac{n^2}{\rho^2} + \frac{\gamma(m^2+n^2)}{2\rho^2}\right) & \left(-\frac{\gamma n}{\rho} + \frac{n}{\rho}\right) & (\gamma-1) & \left(-\frac{\gamma n}{\rho} + \frac{n}{\rho}\right) \\ -\frac{n\gamma e}{\rho^2} + \frac{2(\gamma n(m^2+n^2))}{2\rho^3} - \frac{n(m^2+n^2)}{\rho^3} & \left(\frac{\gamma n m}{\rho^2} + \frac{2mn}{\rho^2}\right) & & \left(\frac{\gamma e}{\rho} + \frac{3\gamma n^2}{2\rho^2} + \frac{3n^2}{2\rho^2}\right) \end{bmatrix}$$

$F = AU = (A^+ + A^-)U = F^+ + F^-$

positive eigenvalues, negative eigenvalues

Flux splitting in non-transformed domain:

$$\therefore \frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F^+ + F^-) + \frac{\partial}{\partial y}(G^+ + G^-) = 0$$

$$\frac{\partial U}{\partial t} + \frac{\partial F^+}{\partial x} + \frac{\partial F^-}{\partial x} + \frac{\partial G^+}{\partial y} + \frac{\partial G^-}{\partial y} = 0$$

$$\frac{U_{ij}^{n+1} - U_{ij}^n}{\Delta t} + \frac{F_{ij}^+ - F_{i-1,j}^+}{\Delta x} + \frac{\hat{F}_{i+1,j}^- - F_{ij}^-}{\Delta x} + \frac{G_{ij}^+ - G_{ij-1}^+}{\Delta y} + \frac{G_{ij+1}^- - G_{ij}^-}{\Delta y} = 0$$

For euler implicit, F^+, F^-, G^+ & G^- are all at time level $n+1$

↳ Similar logic applied on the transformed equations

⇒ Central differencing outside derivatives

$$\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \xi}(J\xi_x F + J\xi_y G) + \frac{\partial}{\partial \eta}(J\eta_x F + J\eta_y G) = 0$$

$$\frac{(J\tau_t U)_{i+\frac{1}{2},j}^n - (J\tau_t U)_{i-\frac{1}{2},j}^n}{\Delta\tau} \qquad \frac{(J\xi_x F + J\xi_y G)_{i+\frac{1}{2},j}^n - (J\xi_x F + J\xi_y G)_{i-\frac{1}{2},j}^n}{\Delta\xi}$$

↳ we can apply this on the transformed equation

(f) Develop an algorithm to generate solutions to the FDE/FVEs that uses approximate factorization.

Algorithm:

1) Discretize domain: $\Delta\eta$, $\Delta\xi$, Δt
   - ↳ choose IL, JL
   - ↳ set $\Delta\eta = \Delta\xi = 1$

2) Solve for Y_1, Y_2, X_1, X_2, $\frac{dY_1}{d\xi}$, $\frac{dY_2}{d\xi}$ & h coefficients

3) Euler implicit time-differencing + linearization

$$\frac{U_{ij}^{n+1} - U_{ij}^n}{\Delta t} = -\left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}\right)_{ij}^{n+1}$$

$$[I + \Delta t(D_x A + D_y B)]\Delta U_{ij}^{n+1} = RHS_{ij}$$

$D_x A$, $D_y B$: from above transformed equations

$$RHS = -\Delta t\left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}\right)_{ij}^n$$

4) Approximate factorization in spatial direction

$$\left\{I + \Delta t\left[D_x(A^+ + A^-) + D_y(B^+ + B^-)\right]\right\}\Delta U_{ij}^{n+1} = RHS_{ij}$$

$$\left[I + \Delta t(L+K)\right]\Delta U_{ij}^{n+1} = RHS_{ij}$$

$$L = S_x^b A^+ + S_y^b B^+ \quad ; \quad K = S_x^f A^- + S_y^f B^-$$

S^b & S^f are backward & forward difference operators

4) The equations are solved by splitting the operator yields (similar to problem 1)

(g) Write a computer program for the algorithm.

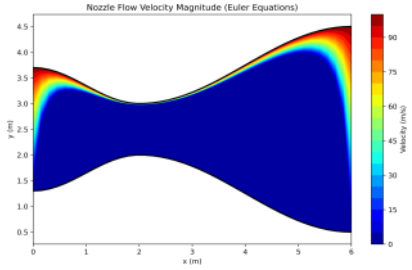
I have tried to program the above algorithm in my code (used LLM for debugging and help) - find attached.

(h) Obtain the steady-state solution

For steady state solutions, LHS can be 1st order accurate but RHS is second order accurate

The steady state solution is when there is not time dependence i.e. the solution is time invariant and the governing equations become just

$$\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$



(i) Discuss accuracy of results with a grid sensitivity study.

There seems to be a bug in the code that I was not able to resolve due to time constraint so my analysis is only limited to the results I obtained. Here are some expected outcomes/analysis based on the physics of converging-diverging nozzle: we expect that the maximum velocity should reach asymptotic value with grid refinement and finer grids should show decreasing differences in the parameters.

Further refining the grid did not change the contour plot solution. Therefore, the observed solution convergence across grid refinements indicates that discretization errors from mesh resolution were negligible. Hence, the remaining numerical errors most likely are from the accuracy of finite difference formulation (assuming no bugs in the code).